

# Final Project

STAT 4610/5610 Spring 2026

2026-03-30

## Final Project: Applying Various Methods to Answer Questions of Interest

### What individuals need to do

1. Given your team's data set and your sub-research question, fit at least three statistical learning models. For each model fit, you should also play around with different tuning parameters, likely using CV.

How is the median house price in \$1000s affected by locational variables such as chas, dis, rad.

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library(ISLR2)
Boston
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|    | crim    | zn   | indus | chas | nox    | rm    | age   | dis    | rad | tax | ptratio | lstat |
|----|---------|------|-------|------|--------|-------|-------|--------|-----|-----|---------|-------|
| 1  | 0.00632 | 18.0 | 2.31  | 0    | 0.5380 | 6.575 | 65.2  | 4.0900 | 1   | 296 | 15.3    | 4.98  |
| 2  | 0.02731 | 0.0  | 7.07  | 0    | 0.4690 | 6.421 | 78.9  | 4.9671 | 2   | 242 | 17.8    | 9.14  |
| 3  | 0.02729 | 0.0  | 7.07  | 0    | 0.4690 | 7.185 | 61.1  | 4.9671 | 2   | 242 | 17.8    | 4.03  |
| 4  | 0.03237 | 0.0  | 2.18  | 0    | 0.4580 | 6.998 | 45.8  | 6.0622 | 3   | 222 | 18.7    | 2.94  |
| 5  | 0.06905 | 0.0  | 2.18  | 0    | 0.4580 | 7.147 | 54.2  | 6.0622 | 3   | 222 | 18.7    | 5.33  |
| 6  | 0.02985 | 0.0  | 2.18  | 0    | 0.4580 | 6.430 | 58.7  | 6.0622 | 3   | 222 | 18.7    | 5.21  |
| 7  | 0.08829 | 12.5 | 7.87  | 0    | 0.5240 | 6.012 | 66.6  | 5.5605 | 5   | 311 | 15.2    | 12.43 |
| 8  | 0.14455 | 12.5 | 7.87  | 0    | 0.5240 | 6.172 | 96.1  | 5.9505 | 5   | 311 | 15.2    | 19.15 |
| 9  | 0.21124 | 12.5 | 7.87  | 0    | 0.5240 | 5.631 | 100.0 | 6.0821 | 5   | 311 | 15.2    | 29.93 |
| 10 | 0.17004 | 12.5 | 7.87  | 0    | 0.5240 | 6.004 | 85.9  | 6.5921 | 5   | 311 | 15.2    | 17.10 |
| 11 | 0.22489 | 12.5 | 7.87  | 0    | 0.5240 | 6.377 | 94.3  | 6.3467 | 5   | 311 | 15.2    | 20.45 |
| 12 | 0.11747 | 12.5 | 7.87  | 0    | 0.5240 | 6.009 | 82.9  | 6.2267 | 5   | 311 | 15.2    | 13.27 |
| 13 | 0.09378 | 12.5 | 7.87  | 0    | 0.5240 | 5.889 | 39.0  | 5.4509 | 5   | 311 | 15.2    | 15.71 |
| 14 | 0.62976 | 0.0  | 8.14  | 0    | 0.5380 | 5.949 | 61.8  | 4.7075 | 4   | 307 | 21.0    | 8.26  |
| 15 | 0.63796 | 0.0  | 8.14  | 0    | 0.5380 | 6.096 | 84.5  | 4.4619 | 4   | 307 | 21.0    | 10.26 |
| 16 | 0.62739 | 0.0  | 8.14  | 0    | 0.5380 | 5.834 | 56.5  | 4.4986 | 4   | 307 | 21.0    | 8.47  |

|    |         |       |      |   |        |       |       |        |   |     |      |       |
|----|---------|-------|------|---|--------|-------|-------|--------|---|-----|------|-------|
| 17 | 1.05393 | 0.0   | 8.14 | 0 | 0.5380 | 5.935 | 29.3  | 4.4986 | 4 | 307 | 21.0 | 6.58  |
| 18 | 0.78420 | 0.0   | 8.14 | 0 | 0.5380 | 5.990 | 81.7  | 4.2579 | 4 | 307 | 21.0 | 14.67 |
| 19 | 0.80271 | 0.0   | 8.14 | 0 | 0.5380 | 5.456 | 36.6  | 3.7965 | 4 | 307 | 21.0 | 11.69 |
| 20 | 0.72580 | 0.0   | 8.14 | 0 | 0.5380 | 5.727 | 69.5  | 3.7965 | 4 | 307 | 21.0 | 11.28 |
| 21 | 1.25179 | 0.0   | 8.14 | 0 | 0.5380 | 5.570 | 98.1  | 3.7979 | 4 | 307 | 21.0 | 21.02 |
| 22 | 0.85204 | 0.0   | 8.14 | 0 | 0.5380 | 5.965 | 89.2  | 4.0123 | 4 | 307 | 21.0 | 13.83 |
| 23 | 1.23247 | 0.0   | 8.14 | 0 | 0.5380 | 6.142 | 91.7  | 3.9769 | 4 | 307 | 21.0 | 18.72 |
| 24 | 0.98843 | 0.0   | 8.14 | 0 | 0.5380 | 5.813 | 100.0 | 4.0952 | 4 | 307 | 21.0 | 19.88 |
| 25 | 0.75026 | 0.0   | 8.14 | 0 | 0.5380 | 5.924 | 94.1  | 4.3996 | 4 | 307 | 21.0 | 16.30 |
| 26 | 0.84054 | 0.0   | 8.14 | 0 | 0.5380 | 5.599 | 85.7  | 4.4546 | 4 | 307 | 21.0 | 16.51 |
| 27 | 0.67191 | 0.0   | 8.14 | 0 | 0.5380 | 5.813 | 90.3  | 4.6820 | 4 | 307 | 21.0 | 14.81 |
| 28 | 0.95577 | 0.0   | 8.14 | 0 | 0.5380 | 6.047 | 88.8  | 4.4534 | 4 | 307 | 21.0 | 17.28 |
| 29 | 0.77299 | 0.0   | 8.14 | 0 | 0.5380 | 6.495 | 94.4  | 4.4547 | 4 | 307 | 21.0 | 12.80 |
| 30 | 1.00245 | 0.0   | 8.14 | 0 | 0.5380 | 6.674 | 87.3  | 4.2390 | 4 | 307 | 21.0 | 11.98 |
| 31 | 1.13081 | 0.0   | 8.14 | 0 | 0.5380 | 5.713 | 94.1  | 4.2330 | 4 | 307 | 21.0 | 22.60 |
| 32 | 1.35472 | 0.0   | 8.14 | 0 | 0.5380 | 6.072 | 100.0 | 4.1750 | 4 | 307 | 21.0 | 13.04 |
| 33 | 1.38799 | 0.0   | 8.14 | 0 | 0.5380 | 5.950 | 82.0  | 3.9900 | 4 | 307 | 21.0 | 27.71 |
| 34 | 1.15172 | 0.0   | 8.14 | 0 | 0.5380 | 5.701 | 95.0  | 3.7872 | 4 | 307 | 21.0 | 18.35 |
| 35 | 1.61282 | 0.0   | 8.14 | 0 | 0.5380 | 6.096 | 96.9  | 3.7598 | 4 | 307 | 21.0 | 20.34 |
| 36 | 0.06417 | 0.0   | 5.96 | 0 | 0.4990 | 5.933 | 68.2  | 3.3603 | 5 | 279 | 19.2 | 9.68  |
| 37 | 0.09744 | 0.0   | 5.96 | 0 | 0.4990 | 5.841 | 61.4  | 3.3779 | 5 | 279 | 19.2 | 11.41 |
| 38 | 0.08014 | 0.0   | 5.96 | 0 | 0.4990 | 5.850 | 41.5  | 3.9342 | 5 | 279 | 19.2 | 8.77  |
| 39 | 0.17505 | 0.0   | 5.96 | 0 | 0.4990 | 5.966 | 30.2  | 3.8473 | 5 | 279 | 19.2 | 10.13 |
| 40 | 0.02763 | 75.0  | 2.95 | 0 | 0.4280 | 6.595 | 21.8  | 5.4011 | 3 | 252 | 18.3 | 4.32  |
| 41 | 0.03359 | 75.0  | 2.95 | 0 | 0.4280 | 7.024 | 15.8  | 5.4011 | 3 | 252 | 18.3 | 1.98  |
| 42 | 0.12744 | 0.0   | 6.91 | 0 | 0.4480 | 6.770 | 2.9   | 5.7209 | 3 | 233 | 17.9 | 4.84  |
| 43 | 0.14150 | 0.0   | 6.91 | 0 | 0.4480 | 6.169 | 6.6   | 5.7209 | 3 | 233 | 17.9 | 5.81  |
| 44 | 0.15936 | 0.0   | 6.91 | 0 | 0.4480 | 6.211 | 6.5   | 5.7209 | 3 | 233 | 17.9 | 7.44  |
| 45 | 0.12269 | 0.0   | 6.91 | 0 | 0.4480 | 6.069 | 40.0  | 5.7209 | 3 | 233 | 17.9 | 9.55  |
| 46 | 0.17142 | 0.0   | 6.91 | 0 | 0.4480 | 5.682 | 33.8  | 5.1004 | 3 | 233 | 17.9 | 10.21 |
| 47 | 0.18836 | 0.0   | 6.91 | 0 | 0.4480 | 5.786 | 33.3  | 5.1004 | 3 | 233 | 17.9 | 14.15 |
| 48 | 0.22927 | 0.0   | 6.91 | 0 | 0.4480 | 6.030 | 85.5  | 5.6894 | 3 | 233 | 17.9 | 18.80 |
| 49 | 0.25387 | 0.0   | 6.91 | 0 | 0.4480 | 5.399 | 95.3  | 5.8700 | 3 | 233 | 17.9 | 30.81 |
| 50 | 0.21977 | 0.0   | 6.91 | 0 | 0.4480 | 5.602 | 62.0  | 6.0877 | 3 | 233 | 17.9 | 16.20 |
| 51 | 0.08873 | 21.0  | 5.64 | 0 | 0.4390 | 5.963 | 45.7  | 6.8147 | 4 | 243 | 16.8 | 13.45 |
| 52 | 0.04337 | 21.0  | 5.64 | 0 | 0.4390 | 6.115 | 63.0  | 6.8147 | 4 | 243 | 16.8 | 9.43  |
| 53 | 0.05360 | 21.0  | 5.64 | 0 | 0.4390 | 6.511 | 21.1  | 6.8147 | 4 | 243 | 16.8 | 5.28  |
| 54 | 0.04981 | 21.0  | 5.64 | 0 | 0.4390 | 5.998 | 21.4  | 6.8147 | 4 | 243 | 16.8 | 8.43  |
| 55 | 0.01360 | 75.0  | 4.00 | 0 | 0.4100 | 5.888 | 47.6  | 7.3197 | 3 | 469 | 21.1 | 14.80 |
| 56 | 0.01311 | 90.0  | 1.22 | 0 | 0.4030 | 7.249 | 21.9  | 8.6966 | 5 | 226 | 17.9 | 4.81  |
| 57 | 0.02055 | 85.0  | 0.74 | 0 | 0.4100 | 6.383 | 35.7  | 9.1876 | 2 | 313 | 17.3 | 5.77  |
| 58 | 0.01432 | 100.0 | 1.32 | 0 | 0.4110 | 6.816 | 40.5  | 8.3248 | 5 | 256 | 15.1 | 3.95  |
| 59 | 0.15445 | 25.0  | 5.13 | 0 | 0.4530 | 6.145 | 29.2  | 7.8148 | 8 | 284 | 19.7 | 6.86  |

|     |         |      |       |   |        |       |      |        |   |     |      |       |
|-----|---------|------|-------|---|--------|-------|------|--------|---|-----|------|-------|
| 60  | 0.10328 | 25.0 | 5.13  | 0 | 0.4530 | 5.927 | 47.2 | 6.9320 | 8 | 284 | 19.7 | 9.22  |
| 61  | 0.14932 | 25.0 | 5.13  | 0 | 0.4530 | 5.741 | 66.2 | 7.2254 | 8 | 284 | 19.7 | 13.15 |
| 62  | 0.17171 | 25.0 | 5.13  | 0 | 0.4530 | 5.966 | 93.4 | 6.8185 | 8 | 284 | 19.7 | 14.44 |
| 63  | 0.11027 | 25.0 | 5.13  | 0 | 0.4530 | 6.456 | 67.8 | 7.2255 | 8 | 284 | 19.7 | 6.73  |
| 64  | 0.12650 | 25.0 | 5.13  | 0 | 0.4530 | 6.762 | 43.4 | 7.9809 | 8 | 284 | 19.7 | 9.50  |
| 65  | 0.01951 | 17.5 | 1.38  | 0 | 0.4161 | 7.104 | 59.5 | 9.2229 | 3 | 216 | 18.6 | 8.05  |
| 66  | 0.03584 | 80.0 | 3.37  | 0 | 0.3980 | 6.290 | 17.8 | 6.6115 | 4 | 337 | 16.1 | 4.67  |
| 67  | 0.04379 | 80.0 | 3.37  | 0 | 0.3980 | 5.787 | 31.1 | 6.6115 | 4 | 337 | 16.1 | 10.24 |
| 68  | 0.05789 | 12.5 | 6.07  | 0 | 0.4090 | 5.878 | 21.4 | 6.4980 | 4 | 345 | 18.9 | 8.10  |
| 69  | 0.13554 | 12.5 | 6.07  | 0 | 0.4090 | 5.594 | 36.8 | 6.4980 | 4 | 345 | 18.9 | 13.09 |
| 70  | 0.12816 | 12.5 | 6.07  | 0 | 0.4090 | 5.885 | 33.0 | 6.4980 | 4 | 345 | 18.9 | 8.79  |
| 71  | 0.08826 | 0.0  | 10.81 | 0 | 0.4130 | 6.417 | 6.6  | 5.2873 | 4 | 305 | 19.2 | 6.72  |
| 72  | 0.15876 | 0.0  | 10.81 | 0 | 0.4130 | 5.961 | 17.5 | 5.2873 | 4 | 305 | 19.2 | 9.88  |
| 73  | 0.09164 | 0.0  | 10.81 | 0 | 0.4130 | 6.065 | 7.8  | 5.2873 | 4 | 305 | 19.2 | 5.52  |
| 74  | 0.19539 | 0.0  | 10.81 | 0 | 0.4130 | 6.245 | 6.2  | 5.2873 | 4 | 305 | 19.2 | 7.54  |
| 75  | 0.07896 | 0.0  | 12.83 | 0 | 0.4370 | 6.273 | 6.0  | 4.2515 | 5 | 398 | 18.7 | 6.78  |
| 76  | 0.09512 | 0.0  | 12.83 | 0 | 0.4370 | 6.286 | 45.0 | 4.5026 | 5 | 398 | 18.7 | 8.94  |
| 77  | 0.10153 | 0.0  | 12.83 | 0 | 0.4370 | 6.279 | 74.5 | 4.0522 | 5 | 398 | 18.7 | 11.97 |
| 78  | 0.08707 | 0.0  | 12.83 | 0 | 0.4370 | 6.140 | 45.8 | 4.0905 | 5 | 398 | 18.7 | 10.27 |
| 79  | 0.05646 | 0.0  | 12.83 | 0 | 0.4370 | 6.232 | 53.7 | 5.0141 | 5 | 398 | 18.7 | 12.34 |
| 80  | 0.08387 | 0.0  | 12.83 | 0 | 0.4370 | 5.874 | 36.6 | 4.5026 | 5 | 398 | 18.7 | 9.10  |
| 81  | 0.04113 | 25.0 | 4.86  | 0 | 0.4260 | 6.727 | 33.5 | 5.4007 | 4 | 281 | 19.0 | 5.29  |
| 82  | 0.04462 | 25.0 | 4.86  | 0 | 0.4260 | 6.619 | 70.4 | 5.4007 | 4 | 281 | 19.0 | 7.22  |
| 83  | 0.03659 | 25.0 | 4.86  | 0 | 0.4260 | 6.302 | 32.2 | 5.4007 | 4 | 281 | 19.0 | 6.72  |
| 84  | 0.03551 | 25.0 | 4.86  | 0 | 0.4260 | 6.167 | 46.7 | 5.4007 | 4 | 281 | 19.0 | 7.51  |
| 85  | 0.05059 | 0.0  | 4.49  | 0 | 0.4490 | 6.389 | 48.0 | 4.7794 | 3 | 247 | 18.5 | 9.62  |
| 86  | 0.05735 | 0.0  | 4.49  | 0 | 0.4490 | 6.630 | 56.1 | 4.4377 | 3 | 247 | 18.5 | 6.53  |
| 87  | 0.05188 | 0.0  | 4.49  | 0 | 0.4490 | 6.015 | 45.1 | 4.4272 | 3 | 247 | 18.5 | 12.86 |
| 88  | 0.07151 | 0.0  | 4.49  | 0 | 0.4490 | 6.121 | 56.8 | 3.7476 | 3 | 247 | 18.5 | 8.44  |
| 89  | 0.05660 | 0.0  | 3.41  | 0 | 0.4890 | 7.007 | 86.3 | 3.4217 | 2 | 270 | 17.8 | 5.50  |
| 90  | 0.05302 | 0.0  | 3.41  | 0 | 0.4890 | 7.079 | 63.1 | 3.4145 | 2 | 270 | 17.8 | 5.70  |
| 91  | 0.04684 | 0.0  | 3.41  | 0 | 0.4890 | 6.417 | 66.1 | 3.0923 | 2 | 270 | 17.8 | 8.81  |
| 92  | 0.03932 | 0.0  | 3.41  | 0 | 0.4890 | 6.405 | 73.9 | 3.0921 | 2 | 270 | 17.8 | 8.20  |
| 93  | 0.04203 | 28.0 | 15.04 | 0 | 0.4640 | 6.442 | 53.6 | 3.6659 | 4 | 270 | 18.2 | 8.16  |
| 94  | 0.02875 | 28.0 | 15.04 | 0 | 0.4640 | 6.211 | 28.9 | 3.6659 | 4 | 270 | 18.2 | 6.21  |
| 95  | 0.04294 | 28.0 | 15.04 | 0 | 0.4640 | 6.249 | 77.3 | 3.6150 | 4 | 270 | 18.2 | 10.59 |
| 96  | 0.12204 | 0.0  | 2.89  | 0 | 0.4450 | 6.625 | 57.8 | 3.4952 | 2 | 276 | 18.0 | 6.65  |
| 97  | 0.11504 | 0.0  | 2.89  | 0 | 0.4450 | 6.163 | 69.6 | 3.4952 | 2 | 276 | 18.0 | 11.34 |
| 98  | 0.12083 | 0.0  | 2.89  | 0 | 0.4450 | 8.069 | 76.0 | 3.4952 | 2 | 276 | 18.0 | 4.21  |
| 99  | 0.08187 | 0.0  | 2.89  | 0 | 0.4450 | 7.820 | 36.9 | 3.4952 | 2 | 276 | 18.0 | 3.57  |
| 100 | 0.06860 | 0.0  | 2.89  | 0 | 0.4450 | 7.416 | 62.5 | 3.4952 | 2 | 276 | 18.0 | 6.19  |
| 101 | 0.14866 | 0.0  | 8.56  | 0 | 0.5200 | 6.727 | 79.9 | 2.7778 | 5 | 384 | 20.9 | 9.42  |
| 102 | 0.11432 | 0.0  | 8.56  | 0 | 0.5200 | 6.781 | 71.3 | 2.8561 | 5 | 384 | 20.9 | 7.67  |

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|-----|---------|-----|-------|---|--------|-------|-------|--------|---|-----|------|-------|
| 103 | 0.22876 | 0.0 | 8.56  | 0 | 0.5200 | 6.405 | 85.4  | 2.7147 | 5 | 384 | 20.9 | 10.63 |
| 104 | 0.21161 | 0.0 | 8.56  | 0 | 0.5200 | 6.137 | 87.4  | 2.7147 | 5 | 384 | 20.9 | 13.44 |
| 105 | 0.13960 | 0.0 | 8.56  | 0 | 0.5200 | 6.167 | 90.0  | 2.4210 | 5 | 384 | 20.9 | 12.33 |
| 106 | 0.13262 | 0.0 | 8.56  | 0 | 0.5200 | 5.851 | 96.7  | 2.1069 | 5 | 384 | 20.9 | 16.47 |
| 107 | 0.17120 | 0.0 | 8.56  | 0 | 0.5200 | 5.836 | 91.9  | 2.2110 | 5 | 384 | 20.9 | 18.66 |
| 108 | 0.13117 | 0.0 | 8.56  | 0 | 0.5200 | 6.127 | 85.2  | 2.1224 | 5 | 384 | 20.9 | 14.09 |
| 109 | 0.12802 | 0.0 | 8.56  | 0 | 0.5200 | 6.474 | 97.1  | 2.4329 | 5 | 384 | 20.9 | 12.27 |
| 110 | 0.26363 | 0.0 | 8.56  | 0 | 0.5200 | 6.229 | 91.2  | 2.5451 | 5 | 384 | 20.9 | 15.55 |
| 111 | 0.10793 | 0.0 | 8.56  | 0 | 0.5200 | 6.195 | 54.4  | 2.7778 | 5 | 384 | 20.9 | 13.00 |
| 112 | 0.10084 | 0.0 | 10.01 | 0 | 0.5470 | 6.715 | 81.6  | 2.6775 | 6 | 432 | 17.8 | 10.16 |
| 113 | 0.12329 | 0.0 | 10.01 | 0 | 0.5470 | 5.913 | 92.9  | 2.3534 | 6 | 432 | 17.8 | 16.21 |
| 114 | 0.22212 | 0.0 | 10.01 | 0 | 0.5470 | 6.092 | 95.4  | 2.5480 | 6 | 432 | 17.8 | 17.09 |
| 115 | 0.14231 | 0.0 | 10.01 | 0 | 0.5470 | 6.254 | 84.2  | 2.2565 | 6 | 432 | 17.8 | 10.45 |
| 116 | 0.17134 | 0.0 | 10.01 | 0 | 0.5470 | 5.928 | 88.2  | 2.4631 | 6 | 432 | 17.8 | 15.76 |
| 117 | 0.13158 | 0.0 | 10.01 | 0 | 0.5470 | 6.176 | 72.5  | 2.7301 | 6 | 432 | 17.8 | 12.04 |
| 118 | 0.15098 | 0.0 | 10.01 | 0 | 0.5470 | 6.021 | 82.6  | 2.7474 | 6 | 432 | 17.8 | 10.30 |
| 119 | 0.13058 | 0.0 | 10.01 | 0 | 0.5470 | 5.872 | 73.1  | 2.4775 | 6 | 432 | 17.8 | 15.37 |
| 120 | 0.14476 | 0.0 | 10.01 | 0 | 0.5470 | 5.731 | 65.2  | 2.7592 | 6 | 432 | 17.8 | 13.61 |
| 121 | 0.06899 | 0.0 | 25.65 | 0 | 0.5810 | 5.870 | 69.7  | 2.2577 | 2 | 188 | 19.1 | 14.37 |
| 122 | 0.07165 | 0.0 | 25.65 | 0 | 0.5810 | 6.004 | 84.1  | 2.1974 | 2 | 188 | 19.1 | 14.27 |
| 123 | 0.09299 | 0.0 | 25.65 | 0 | 0.5810 | 5.961 | 92.9  | 2.0869 | 2 | 188 | 19.1 | 17.93 |
| 124 | 0.15038 | 0.0 | 25.65 | 0 | 0.5810 | 5.856 | 97.0  | 1.9444 | 2 | 188 | 19.1 | 25.41 |
| 125 | 0.09849 | 0.0 | 25.65 | 0 | 0.5810 | 5.879 | 95.8  | 2.0063 | 2 | 188 | 19.1 | 17.58 |
| 126 | 0.16902 | 0.0 | 25.65 | 0 | 0.5810 | 5.986 | 88.4  | 1.9929 | 2 | 188 | 19.1 | 14.81 |
| 127 | 0.38735 | 0.0 | 25.65 | 0 | 0.5810 | 5.613 | 95.6  | 1.7572 | 2 | 188 | 19.1 | 27.26 |
| 128 | 0.25915 | 0.0 | 21.89 | 0 | 0.6240 | 5.693 | 96.0  | 1.7883 | 4 | 437 | 21.2 | 17.19 |
| 129 | 0.32543 | 0.0 | 21.89 | 0 | 0.6240 | 6.431 | 98.8  | 1.8125 | 4 | 437 | 21.2 | 15.39 |
| 130 | 0.88125 | 0.0 | 21.89 | 0 | 0.6240 | 5.637 | 94.7  | 1.9799 | 4 | 437 | 21.2 | 18.34 |
| 131 | 0.34006 | 0.0 | 21.89 | 0 | 0.6240 | 6.458 | 98.9  | 2.1185 | 4 | 437 | 21.2 | 12.60 |
| 132 | 1.19294 | 0.0 | 21.89 | 0 | 0.6240 | 6.326 | 97.7  | 2.2710 | 4 | 437 | 21.2 | 12.26 |
| 133 | 0.59005 | 0.0 | 21.89 | 0 | 0.6240 | 6.372 | 97.9  | 2.3274 | 4 | 437 | 21.2 | 11.12 |
| 134 | 0.32982 | 0.0 | 21.89 | 0 | 0.6240 | 5.822 | 95.4  | 2.4699 | 4 | 437 | 21.2 | 15.03 |
| 135 | 0.97617 | 0.0 | 21.89 | 0 | 0.6240 | 5.757 | 98.4  | 2.3460 | 4 | 437 | 21.2 | 17.31 |
| 136 | 0.55778 | 0.0 | 21.89 | 0 | 0.6240 | 6.335 | 98.2  | 2.1107 | 4 | 437 | 21.2 | 16.96 |
| 137 | 0.32264 | 0.0 | 21.89 | 0 | 0.6240 | 5.942 | 93.5  | 1.9669 | 4 | 437 | 21.2 | 16.90 |
| 138 | 0.35233 | 0.0 | 21.89 | 0 | 0.6240 | 6.454 | 98.4  | 1.8498 | 4 | 437 | 21.2 | 14.59 |
| 139 | 0.24980 | 0.0 | 21.89 | 0 | 0.6240 | 5.857 | 98.2  | 1.6686 | 4 | 437 | 21.2 | 21.32 |
| 140 | 0.54452 | 0.0 | 21.89 | 0 | 0.6240 | 6.151 | 97.9  | 1.6687 | 4 | 437 | 21.2 | 18.46 |
| 141 | 0.29090 | 0.0 | 21.89 | 0 | 0.6240 | 6.174 | 93.6  | 1.6119 | 4 | 437 | 21.2 | 24.16 |
| 142 | 1.62864 | 0.0 | 21.89 | 0 | 0.6240 | 5.019 | 100.0 | 1.4394 | 4 | 437 | 21.2 | 34.41 |
| 143 | 3.32105 | 0.0 | 19.58 | 1 | 0.8710 | 5.403 | 100.0 | 1.3216 | 5 | 403 | 14.7 | 26.82 |
| 144 | 4.09740 | 0.0 | 19.58 | 0 | 0.8710 | 5.468 | 100.0 | 1.4118 | 5 | 403 | 14.7 | 26.42 |
| 145 | 2.77974 | 0.0 | 19.58 | 0 | 0.8710 | 4.903 | 97.8  | 1.3459 | 5 | 403 | 14.7 | 29.29 |

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|-----|---------|------|-------|---|--------|-------|-------|--------|---|-----|------|-------|
| 146 | 2.37934 | 0.0  | 19.58 | 0 | 0.8710 | 6.130 | 100.0 | 1.4191 | 5 | 403 | 14.7 | 27.80 |
| 147 | 2.15505 | 0.0  | 19.58 | 0 | 0.8710 | 5.628 | 100.0 | 1.5166 | 5 | 403 | 14.7 | 16.65 |
| 148 | 2.36862 | 0.0  | 19.58 | 0 | 0.8710 | 4.926 | 95.7  | 1.4608 | 5 | 403 | 14.7 | 29.53 |
| 149 | 2.33099 | 0.0  | 19.58 | 0 | 0.8710 | 5.186 | 93.8  | 1.5296 | 5 | 403 | 14.7 | 28.32 |
| 150 | 2.73397 | 0.0  | 19.58 | 0 | 0.8710 | 5.597 | 94.9  | 1.5257 | 5 | 403 | 14.7 | 21.45 |
| 151 | 1.65660 | 0.0  | 19.58 | 0 | 0.8710 | 6.122 | 97.3  | 1.6180 | 5 | 403 | 14.7 | 14.10 |
| 152 | 1.49632 | 0.0  | 19.58 | 0 | 0.8710 | 5.404 | 100.0 | 1.5916 | 5 | 403 | 14.7 | 13.28 |
| 153 | 1.12658 | 0.0  | 19.58 | 1 | 0.8710 | 5.012 | 88.0  | 1.6102 | 5 | 403 | 14.7 | 12.12 |
| 154 | 2.14918 | 0.0  | 19.58 | 0 | 0.8710 | 5.709 | 98.5  | 1.6232 | 5 | 403 | 14.7 | 15.79 |
| 155 | 1.41385 | 0.0  | 19.58 | 1 | 0.8710 | 6.129 | 96.0  | 1.7494 | 5 | 403 | 14.7 | 15.12 |
| 156 | 3.53501 | 0.0  | 19.58 | 1 | 0.8710 | 6.152 | 82.6  | 1.7455 | 5 | 403 | 14.7 | 15.02 |
| 157 | 2.44668 | 0.0  | 19.58 | 0 | 0.8710 | 5.272 | 94.0  | 1.7364 | 5 | 403 | 14.7 | 16.14 |
| 158 | 1.22358 | 0.0  | 19.58 | 0 | 0.6050 | 6.943 | 97.4  | 1.8773 | 5 | 403 | 14.7 | 4.59  |
| 159 | 1.34284 | 0.0  | 19.58 | 0 | 0.6050 | 6.066 | 100.0 | 1.7573 | 5 | 403 | 14.7 | 6.43  |
| 160 | 1.42502 | 0.0  | 19.58 | 0 | 0.8710 | 6.510 | 100.0 | 1.7659 | 5 | 403 | 14.7 | 7.39  |
| 161 | 1.27346 | 0.0  | 19.58 | 1 | 0.6050 | 6.250 | 92.6  | 1.7984 | 5 | 403 | 14.7 | 5.50  |
| 162 | 1.46336 | 0.0  | 19.58 | 0 | 0.6050 | 7.489 | 90.8  | 1.9709 | 5 | 403 | 14.7 | 1.73  |
| 163 | 1.83377 | 0.0  | 19.58 | 1 | 0.6050 | 7.802 | 98.2  | 2.0407 | 5 | 403 | 14.7 | 1.92  |
| 164 | 1.51902 | 0.0  | 19.58 | 1 | 0.6050 | 8.375 | 93.9  | 2.1620 | 5 | 403 | 14.7 | 3.32  |
| 165 | 2.24236 | 0.0  | 19.58 | 0 | 0.6050 | 5.854 | 91.8  | 2.4220 | 5 | 403 | 14.7 | 11.64 |
| 166 | 2.92400 | 0.0  | 19.58 | 0 | 0.6050 | 6.101 | 93.0  | 2.2834 | 5 | 403 | 14.7 | 9.81  |
| 167 | 2.01019 | 0.0  | 19.58 | 0 | 0.6050 | 7.929 | 96.2  | 2.0459 | 5 | 403 | 14.7 | 3.70  |
| 168 | 1.80028 | 0.0  | 19.58 | 0 | 0.6050 | 5.877 | 79.2  | 2.4259 | 5 | 403 | 14.7 | 12.14 |
| 169 | 2.30040 | 0.0  | 19.58 | 0 | 0.6050 | 6.319 | 96.1  | 2.1000 | 5 | 403 | 14.7 | 11.10 |
| 170 | 2.44953 | 0.0  | 19.58 | 0 | 0.6050 | 6.402 | 95.2  | 2.2625 | 5 | 403 | 14.7 | 11.32 |
| 171 | 1.20742 | 0.0  | 19.58 | 0 | 0.6050 | 5.875 | 94.6  | 2.4259 | 5 | 403 | 14.7 | 14.43 |
| 172 | 2.31390 | 0.0  | 19.58 | 0 | 0.6050 | 5.880 | 97.3  | 2.3887 | 5 | 403 | 14.7 | 12.03 |
| 173 | 0.13914 | 0.0  | 4.05  | 0 | 0.5100 | 5.572 | 88.5  | 2.5961 | 5 | 296 | 16.6 | 14.69 |
| 174 | 0.09178 | 0.0  | 4.05  | 0 | 0.5100 | 6.416 | 84.1  | 2.6463 | 5 | 296 | 16.6 | 9.04  |
| 175 | 0.08447 | 0.0  | 4.05  | 0 | 0.5100 | 5.859 | 68.7  | 2.7019 | 5 | 296 | 16.6 | 9.64  |
| 176 | 0.06664 | 0.0  | 4.05  | 0 | 0.5100 | 6.546 | 33.1  | 3.1323 | 5 | 296 | 16.6 | 5.33  |
| 177 | 0.07022 | 0.0  | 4.05  | 0 | 0.5100 | 6.020 | 47.2  | 3.5549 | 5 | 296 | 16.6 | 10.11 |
| 178 | 0.05425 | 0.0  | 4.05  | 0 | 0.5100 | 6.315 | 73.4  | 3.3175 | 5 | 296 | 16.6 | 6.29  |
| 179 | 0.06642 | 0.0  | 4.05  | 0 | 0.5100 | 6.860 | 74.4  | 2.9153 | 5 | 296 | 16.6 | 6.92  |
| 180 | 0.05780 | 0.0  | 2.46  | 0 | 0.4880 | 6.980 | 58.4  | 2.8290 | 3 | 193 | 17.8 | 5.04  |
| 181 | 0.06588 | 0.0  | 2.46  | 0 | 0.4880 | 7.765 | 83.3  | 2.7410 | 3 | 193 | 17.8 | 7.56  |
| 182 | 0.06888 | 0.0  | 2.46  | 0 | 0.4880 | 6.144 | 62.2  | 2.5979 | 3 | 193 | 17.8 | 9.45  |
| 183 | 0.09103 | 0.0  | 2.46  | 0 | 0.4880 | 7.155 | 92.2  | 2.7006 | 3 | 193 | 17.8 | 4.82  |
| 184 | 0.10008 | 0.0  | 2.46  | 0 | 0.4880 | 6.563 | 95.6  | 2.8470 | 3 | 193 | 17.8 | 5.68  |
| 185 | 0.08308 | 0.0  | 2.46  | 0 | 0.4880 | 5.604 | 89.8  | 2.9879 | 3 | 193 | 17.8 | 13.98 |
| 186 | 0.06047 | 0.0  | 2.46  | 0 | 0.4880 | 6.153 | 68.8  | 3.2797 | 3 | 193 | 17.8 | 13.15 |
| 187 | 0.05602 | 0.0  | 2.46  | 0 | 0.4880 | 7.831 | 53.6  | 3.1992 | 3 | 193 | 17.8 | 4.45  |
| 188 | 0.07875 | 45.0 | 3.44  | 0 | 0.4370 | 6.782 | 41.1  | 3.7886 | 5 | 398 | 15.2 | 6.68  |

|     |         |      |       |   |        |       |       |        |   |     |      |       |
|-----|---------|------|-------|---|--------|-------|-------|--------|---|-----|------|-------|
| 189 | 0.12579 | 45.0 | 3.44  | 0 | 0.4370 | 6.556 | 29.1  | 4.5667 | 5 | 398 | 15.2 | 4.56  |
| 190 | 0.08370 | 45.0 | 3.44  | 0 | 0.4370 | 7.185 | 38.9  | 4.5667 | 5 | 398 | 15.2 | 5.39  |
| 191 | 0.09068 | 45.0 | 3.44  | 0 | 0.4370 | 6.951 | 21.5  | 6.4798 | 5 | 398 | 15.2 | 5.10  |
| 192 | 0.06911 | 45.0 | 3.44  | 0 | 0.4370 | 6.739 | 30.8  | 6.4798 | 5 | 398 | 15.2 | 4.69  |
| 193 | 0.08664 | 45.0 | 3.44  | 0 | 0.4370 | 7.178 | 26.3  | 6.4798 | 5 | 398 | 15.2 | 2.87  |
| 194 | 0.02187 | 60.0 | 2.93  | 0 | 0.4010 | 6.800 | 9.9   | 6.2196 | 1 | 265 | 15.6 | 5.03  |
| 195 | 0.01439 | 60.0 | 2.93  | 0 | 0.4010 | 6.604 | 18.8  | 6.2196 | 1 | 265 | 15.6 | 4.38  |
| 196 | 0.01381 | 80.0 | 0.46  | 0 | 0.4220 | 7.875 | 32.0  | 5.6484 | 4 | 255 | 14.4 | 2.97  |
| 197 | 0.04011 | 80.0 | 1.52  | 0 | 0.4040 | 7.287 | 34.1  | 7.3090 | 2 | 329 | 12.6 | 4.08  |
| 198 | 0.04666 | 80.0 | 1.52  | 0 | 0.4040 | 7.107 | 36.6  | 7.3090 | 2 | 329 | 12.6 | 8.61  |
| 199 | 0.03768 | 80.0 | 1.52  | 0 | 0.4040 | 7.274 | 38.3  | 7.3090 | 2 | 329 | 12.6 | 6.62  |
| 200 | 0.03150 | 95.0 | 1.47  | 0 | 0.4030 | 6.975 | 15.3  | 7.6534 | 3 | 402 | 17.0 | 4.56  |
| 201 | 0.01778 | 95.0 | 1.47  | 0 | 0.4030 | 7.135 | 13.9  | 7.6534 | 3 | 402 | 17.0 | 4.45  |
| 202 | 0.03445 | 82.5 | 2.03  | 0 | 0.4150 | 6.162 | 38.4  | 6.2700 | 2 | 348 | 14.7 | 7.43  |
| 203 | 0.02177 | 82.5 | 2.03  | 0 | 0.4150 | 7.610 | 15.7  | 6.2700 | 2 | 348 | 14.7 | 3.11  |
| 204 | 0.03510 | 95.0 | 2.68  | 0 | 0.4161 | 7.853 | 33.2  | 5.1180 | 4 | 224 | 14.7 | 3.81  |
| 205 | 0.02009 | 95.0 | 2.68  | 0 | 0.4161 | 8.034 | 31.9  | 5.1180 | 4 | 224 | 14.7 | 2.88  |
| 206 | 0.13642 | 0.0  | 10.59 | 0 | 0.4890 | 5.891 | 22.3  | 3.9454 | 4 | 277 | 18.6 | 10.87 |
| 207 | 0.22969 | 0.0  | 10.59 | 0 | 0.4890 | 6.326 | 52.5  | 4.3549 | 4 | 277 | 18.6 | 10.97 |
| 208 | 0.25199 | 0.0  | 10.59 | 0 | 0.4890 | 5.783 | 72.7  | 4.3549 | 4 | 277 | 18.6 | 18.06 |
| 209 | 0.13587 | 0.0  | 10.59 | 1 | 0.4890 | 6.064 | 59.1  | 4.2392 | 4 | 277 | 18.6 | 14.66 |
| 210 | 0.43571 | 0.0  | 10.59 | 1 | 0.4890 | 5.344 | 100.0 | 3.8750 | 4 | 277 | 18.6 | 23.09 |
| 211 | 0.17446 | 0.0  | 10.59 | 1 | 0.4890 | 5.960 | 92.1  | 3.8771 | 4 | 277 | 18.6 | 17.27 |
| 212 | 0.37578 | 0.0  | 10.59 | 1 | 0.4890 | 5.404 | 88.6  | 3.6650 | 4 | 277 | 18.6 | 23.98 |
| 213 | 0.21719 | 0.0  | 10.59 | 1 | 0.4890 | 5.807 | 53.8  | 3.6526 | 4 | 277 | 18.6 | 16.03 |
| 214 | 0.14052 | 0.0  | 10.59 | 0 | 0.4890 | 6.375 | 32.3  | 3.9454 | 4 | 277 | 18.6 | 9.38  |
| 215 | 0.28955 | 0.0  | 10.59 | 0 | 0.4890 | 5.412 | 9.8   | 3.5875 | 4 | 277 | 18.6 | 29.55 |
| 216 | 0.19802 | 0.0  | 10.59 | 0 | 0.4890 | 6.182 | 42.4  | 3.9454 | 4 | 277 | 18.6 | 9.47  |
| 217 | 0.04560 | 0.0  | 13.89 | 1 | 0.5500 | 5.888 | 56.0  | 3.1121 | 5 | 276 | 16.4 | 13.51 |
| 218 | 0.07013 | 0.0  | 13.89 | 0 | 0.5500 | 6.642 | 85.1  | 3.4211 | 5 | 276 | 16.4 | 9.69  |
| 219 | 0.11069 | 0.0  | 13.89 | 1 | 0.5500 | 5.951 | 93.8  | 2.8893 | 5 | 276 | 16.4 | 17.92 |
| 220 | 0.11425 | 0.0  | 13.89 | 1 | 0.5500 | 6.373 | 92.4  | 3.3633 | 5 | 276 | 16.4 | 10.50 |
| 221 | 0.35809 | 0.0  | 6.20  | 1 | 0.5070 | 6.951 | 88.5  | 2.8617 | 8 | 307 | 17.4 | 9.71  |
| 222 | 0.40771 | 0.0  | 6.20  | 1 | 0.5070 | 6.164 | 91.3  | 3.0480 | 8 | 307 | 17.4 | 21.46 |
| 223 | 0.62356 | 0.0  | 6.20  | 1 | 0.5070 | 6.879 | 77.7  | 3.2721 | 8 | 307 | 17.4 | 9.93  |
| 224 | 0.61470 | 0.0  | 6.20  | 0 | 0.5070 | 6.618 | 80.8  | 3.2721 | 8 | 307 | 17.4 | 7.60  |
| 225 | 0.31533 | 0.0  | 6.20  | 0 | 0.5040 | 8.266 | 78.3  | 2.8944 | 8 | 307 | 17.4 | 4.14  |
| 226 | 0.52693 | 0.0  | 6.20  | 0 | 0.5040 | 8.725 | 83.0  | 2.8944 | 8 | 307 | 17.4 | 4.63  |
| 227 | 0.38214 | 0.0  | 6.20  | 0 | 0.5040 | 8.040 | 86.5  | 3.2157 | 8 | 307 | 17.4 | 3.13  |
| 228 | 0.41238 | 0.0  | 6.20  | 0 | 0.5040 | 7.163 | 79.9  | 3.2157 | 8 | 307 | 17.4 | 6.36  |
| 229 | 0.29819 | 0.0  | 6.20  | 0 | 0.5040 | 7.686 | 17.0  | 3.3751 | 8 | 307 | 17.4 | 3.92  |
| 230 | 0.44178 | 0.0  | 6.20  | 0 | 0.5040 | 6.552 | 21.4  | 3.3751 | 8 | 307 | 17.4 | 3.76  |
| 231 | 0.53700 | 0.0  | 6.20  | 0 | 0.5040 | 5.981 | 68.1  | 3.6715 | 8 | 307 | 17.4 | 11.65 |

|     |         |      |      |   |        |       |       |        |   |     |      |       |
|-----|---------|------|------|---|--------|-------|-------|--------|---|-----|------|-------|
| 232 | 0.46296 | 0.0  | 6.20 | 0 | 0.5040 | 7.412 | 76.9  | 3.6715 | 8 | 307 | 17.4 | 5.25  |
| 233 | 0.57529 | 0.0  | 6.20 | 0 | 0.5070 | 8.337 | 73.3  | 3.8384 | 8 | 307 | 17.4 | 2.47  |
| 234 | 0.33147 | 0.0  | 6.20 | 0 | 0.5070 | 8.247 | 70.4  | 3.6519 | 8 | 307 | 17.4 | 3.95  |
| 235 | 0.44791 | 0.0  | 6.20 | 1 | 0.5070 | 6.726 | 66.5  | 3.6519 | 8 | 307 | 17.4 | 8.05  |
| 236 | 0.33045 | 0.0  | 6.20 | 0 | 0.5070 | 6.086 | 61.5  | 3.6519 | 8 | 307 | 17.4 | 10.88 |
| 237 | 0.52058 | 0.0  | 6.20 | 1 | 0.5070 | 6.631 | 76.5  | 4.1480 | 8 | 307 | 17.4 | 9.54  |
| 238 | 0.51183 | 0.0  | 6.20 | 0 | 0.5070 | 7.358 | 71.6  | 4.1480 | 8 | 307 | 17.4 | 4.73  |
| 239 | 0.08244 | 30.0 | 4.93 | 0 | 0.4280 | 6.481 | 18.5  | 6.1899 | 6 | 300 | 16.6 | 6.36  |
| 240 | 0.09252 | 30.0 | 4.93 | 0 | 0.4280 | 6.606 | 42.2  | 6.1899 | 6 | 300 | 16.6 | 7.37  |
| 241 | 0.11329 | 30.0 | 4.93 | 0 | 0.4280 | 6.897 | 54.3  | 6.3361 | 6 | 300 | 16.6 | 11.38 |
| 242 | 0.10612 | 30.0 | 4.93 | 0 | 0.4280 | 6.095 | 65.1  | 6.3361 | 6 | 300 | 16.6 | 12.40 |
| 243 | 0.10290 | 30.0 | 4.93 | 0 | 0.4280 | 6.358 | 52.9  | 7.0355 | 6 | 300 | 16.6 | 11.22 |
| 244 | 0.12757 | 30.0 | 4.93 | 0 | 0.4280 | 6.393 | 7.8   | 7.0355 | 6 | 300 | 16.6 | 5.19  |
| 245 | 0.20608 | 22.0 | 5.86 | 0 | 0.4310 | 5.593 | 76.5  | 7.9549 | 7 | 330 | 19.1 | 12.50 |
| 246 | 0.19133 | 22.0 | 5.86 | 0 | 0.4310 | 5.605 | 70.2  | 7.9549 | 7 | 330 | 19.1 | 18.46 |
| 247 | 0.33983 | 22.0 | 5.86 | 0 | 0.4310 | 6.108 | 34.9  | 8.0555 | 7 | 330 | 19.1 | 9.16  |
| 248 | 0.19657 | 22.0 | 5.86 | 0 | 0.4310 | 6.226 | 79.2  | 8.0555 | 7 | 330 | 19.1 | 10.15 |
| 249 | 0.16439 | 22.0 | 5.86 | 0 | 0.4310 | 6.433 | 49.1  | 7.8265 | 7 | 330 | 19.1 | 9.52  |
| 250 | 0.19073 | 22.0 | 5.86 | 0 | 0.4310 | 6.718 | 17.5  | 7.8265 | 7 | 330 | 19.1 | 6.56  |
| 251 | 0.14030 | 22.0 | 5.86 | 0 | 0.4310 | 6.487 | 13.0  | 7.3967 | 7 | 330 | 19.1 | 5.90  |
| 252 | 0.21409 | 22.0 | 5.86 | 0 | 0.4310 | 6.438 | 8.9   | 7.3967 | 7 | 330 | 19.1 | 3.59  |
| 253 | 0.08221 | 22.0 | 5.86 | 0 | 0.4310 | 6.957 | 6.8   | 8.9067 | 7 | 330 | 19.1 | 3.53  |
| 254 | 0.36894 | 22.0 | 5.86 | 0 | 0.4310 | 8.259 | 8.4   | 8.9067 | 7 | 330 | 19.1 | 3.54  |
| 255 | 0.04819 | 80.0 | 3.64 | 0 | 0.3920 | 6.108 | 32.0  | 9.2203 | 1 | 315 | 16.4 | 6.57  |
| 256 | 0.03548 | 80.0 | 3.64 | 0 | 0.3920 | 5.876 | 19.1  | 9.2203 | 1 | 315 | 16.4 | 9.25  |
| 257 | 0.01538 | 90.0 | 3.75 | 0 | 0.3940 | 7.454 | 34.2  | 6.3361 | 3 | 244 | 15.9 | 3.11  |
| 258 | 0.61154 | 20.0 | 3.97 | 0 | 0.6470 | 8.704 | 86.9  | 1.8010 | 5 | 264 | 13.0 | 5.12  |
| 259 | 0.66351 | 20.0 | 3.97 | 0 | 0.6470 | 7.333 | 100.0 | 1.8946 | 5 | 264 | 13.0 | 7.79  |
| 260 | 0.65665 | 20.0 | 3.97 | 0 | 0.6470 | 6.842 | 100.0 | 2.0107 | 5 | 264 | 13.0 | 6.90  |
| 261 | 0.54011 | 20.0 | 3.97 | 0 | 0.6470 | 7.203 | 81.8  | 2.1121 | 5 | 264 | 13.0 | 9.59  |
| 262 | 0.53412 | 20.0 | 3.97 | 0 | 0.6470 | 7.520 | 89.4  | 2.1398 | 5 | 264 | 13.0 | 7.26  |
| 263 | 0.52014 | 20.0 | 3.97 | 0 | 0.6470 | 8.398 | 91.5  | 2.2885 | 5 | 264 | 13.0 | 5.91  |
| 264 | 0.82526 | 20.0 | 3.97 | 0 | 0.6470 | 7.327 | 94.5  | 2.0788 | 5 | 264 | 13.0 | 11.25 |
| 265 | 0.55007 | 20.0 | 3.97 | 0 | 0.6470 | 7.206 | 91.6  | 1.9301 | 5 | 264 | 13.0 | 8.10  |
| 266 | 0.76162 | 20.0 | 3.97 | 0 | 0.6470 | 5.560 | 62.8  | 1.9865 | 5 | 264 | 13.0 | 10.45 |
| 267 | 0.78570 | 20.0 | 3.97 | 0 | 0.6470 | 7.014 | 84.6  | 2.1329 | 5 | 264 | 13.0 | 14.79 |
| 268 | 0.57834 | 20.0 | 3.97 | 0 | 0.5750 | 8.297 | 67.0  | 2.4216 | 5 | 264 | 13.0 | 7.44  |
| 269 | 0.54050 | 20.0 | 3.97 | 0 | 0.5750 | 7.470 | 52.6  | 2.8720 | 5 | 264 | 13.0 | 3.16  |
| 270 | 0.09065 | 20.0 | 6.96 | 1 | 0.4640 | 5.920 | 61.5  | 3.9175 | 3 | 223 | 18.6 | 13.65 |
| 271 | 0.29916 | 20.0 | 6.96 | 0 | 0.4640 | 5.856 | 42.1  | 4.4290 | 3 | 223 | 18.6 | 13.00 |
| 272 | 0.16211 | 20.0 | 6.96 | 0 | 0.4640 | 6.240 | 16.3  | 4.4290 | 3 | 223 | 18.6 | 6.59  |
| 273 | 0.11460 | 20.0 | 6.96 | 0 | 0.4640 | 6.538 | 58.7  | 3.9175 | 3 | 223 | 18.6 | 7.73  |
| 274 | 0.22188 | 20.0 | 6.96 | 1 | 0.4640 | 7.691 | 51.8  | 4.3665 | 3 | 223 | 18.6 | 6.58  |

|     |         |      |       |   |        |       |      |        |   |     |      |       |
|-----|---------|------|-------|---|--------|-------|------|--------|---|-----|------|-------|
| 275 | 0.05644 | 40.0 | 6.41  | 1 | 0.4470 | 6.758 | 32.9 | 4.0776 | 4 | 254 | 17.6 | 3.53  |
| 276 | 0.09604 | 40.0 | 6.41  | 0 | 0.4470 | 6.854 | 42.8 | 4.2673 | 4 | 254 | 17.6 | 2.98  |
| 277 | 0.10469 | 40.0 | 6.41  | 1 | 0.4470 | 7.267 | 49.0 | 4.7872 | 4 | 254 | 17.6 | 6.05  |
| 278 | 0.06127 | 40.0 | 6.41  | 1 | 0.4470 | 6.826 | 27.6 | 4.8628 | 4 | 254 | 17.6 | 4.16  |
| 279 | 0.07978 | 40.0 | 6.41  | 0 | 0.4470 | 6.482 | 32.1 | 4.1403 | 4 | 254 | 17.6 | 7.19  |
| 280 | 0.21038 | 20.0 | 3.33  | 0 | 0.4429 | 6.812 | 32.2 | 4.1007 | 5 | 216 | 14.9 | 4.85  |
| 281 | 0.03578 | 20.0 | 3.33  | 0 | 0.4429 | 7.820 | 64.5 | 4.6947 | 5 | 216 | 14.9 | 3.76  |
| 282 | 0.03705 | 20.0 | 3.33  | 0 | 0.4429 | 6.968 | 37.2 | 5.2447 | 5 | 216 | 14.9 | 4.59  |
| 283 | 0.06129 | 20.0 | 3.33  | 1 | 0.4429 | 7.645 | 49.7 | 5.2119 | 5 | 216 | 14.9 | 3.01  |
| 284 | 0.01501 | 90.0 | 1.21  | 1 | 0.4010 | 7.923 | 24.8 | 5.8850 | 1 | 198 | 13.6 | 3.16  |
| 285 | 0.00906 | 90.0 | 2.97  | 0 | 0.4000 | 7.088 | 20.8 | 7.3073 | 1 | 285 | 15.3 | 7.85  |
| 286 | 0.01096 | 55.0 | 2.25  | 0 | 0.3890 | 6.453 | 31.9 | 7.3073 | 1 | 300 | 15.3 | 8.23  |
| 287 | 0.01965 | 80.0 | 1.76  | 0 | 0.3850 | 6.230 | 31.5 | 9.0892 | 1 | 241 | 18.2 | 12.93 |
| 288 | 0.03871 | 52.5 | 5.32  | 0 | 0.4050 | 6.209 | 31.3 | 7.3172 | 6 | 293 | 16.6 | 7.14  |
| 289 | 0.04590 | 52.5 | 5.32  | 0 | 0.4050 | 6.315 | 45.6 | 7.3172 | 6 | 293 | 16.6 | 7.60  |
| 290 | 0.04297 | 52.5 | 5.32  | 0 | 0.4050 | 6.565 | 22.9 | 7.3172 | 6 | 293 | 16.6 | 9.51  |
| 291 | 0.03502 | 80.0 | 4.95  | 0 | 0.4110 | 6.861 | 27.9 | 5.1167 | 4 | 245 | 19.2 | 3.33  |
| 292 | 0.07886 | 80.0 | 4.95  | 0 | 0.4110 | 7.148 | 27.7 | 5.1167 | 4 | 245 | 19.2 | 3.56  |
| 293 | 0.03615 | 80.0 | 4.95  | 0 | 0.4110 | 6.630 | 23.4 | 5.1167 | 4 | 245 | 19.2 | 4.70  |
| 294 | 0.08265 | 0.0  | 13.92 | 0 | 0.4370 | 6.127 | 18.4 | 5.5027 | 4 | 289 | 16.0 | 8.58  |
| 295 | 0.08199 | 0.0  | 13.92 | 0 | 0.4370 | 6.009 | 42.3 | 5.5027 | 4 | 289 | 16.0 | 10.40 |
| 296 | 0.12932 | 0.0  | 13.92 | 0 | 0.4370 | 6.678 | 31.1 | 5.9604 | 4 | 289 | 16.0 | 6.27  |
| 297 | 0.05372 | 0.0  | 13.92 | 0 | 0.4370 | 6.549 | 51.0 | 5.9604 | 4 | 289 | 16.0 | 7.39  |
| 298 | 0.14103 | 0.0  | 13.92 | 0 | 0.4370 | 5.790 | 58.0 | 6.3200 | 4 | 289 | 16.0 | 15.84 |
| 299 | 0.06466 | 70.0 | 2.24  | 0 | 0.4000 | 6.345 | 20.1 | 7.8278 | 5 | 358 | 14.8 | 4.97  |
| 300 | 0.05561 | 70.0 | 2.24  | 0 | 0.4000 | 7.041 | 10.0 | 7.8278 | 5 | 358 | 14.8 | 4.74  |
| 301 | 0.04417 | 70.0 | 2.24  | 0 | 0.4000 | 6.871 | 47.4 | 7.8278 | 5 | 358 | 14.8 | 6.07  |
| 302 | 0.03537 | 34.0 | 6.09  | 0 | 0.4330 | 6.590 | 40.4 | 5.4917 | 7 | 329 | 16.1 | 9.50  |
| 303 | 0.09266 | 34.0 | 6.09  | 0 | 0.4330 | 6.495 | 18.4 | 5.4917 | 7 | 329 | 16.1 | 8.67  |
| 304 | 0.10000 | 34.0 | 6.09  | 0 | 0.4330 | 6.982 | 17.7 | 5.4917 | 7 | 329 | 16.1 | 4.86  |
| 305 | 0.05515 | 33.0 | 2.18  | 0 | 0.4720 | 7.236 | 41.1 | 4.0220 | 7 | 222 | 18.4 | 6.93  |
| 306 | 0.05479 | 33.0 | 2.18  | 0 | 0.4720 | 6.616 | 58.1 | 3.3700 | 7 | 222 | 18.4 | 8.93  |
| 307 | 0.07503 | 33.0 | 2.18  | 0 | 0.4720 | 7.420 | 71.9 | 3.0992 | 7 | 222 | 18.4 | 6.47  |
| 308 | 0.04932 | 33.0 | 2.18  | 0 | 0.4720 | 6.849 | 70.3 | 3.1827 | 7 | 222 | 18.4 | 7.53  |
| 309 | 0.49298 | 0.0  | 9.90  | 0 | 0.5440 | 6.635 | 82.5 | 3.3175 | 4 | 304 | 18.4 | 4.54  |
| 310 | 0.34940 | 0.0  | 9.90  | 0 | 0.5440 | 5.972 | 76.7 | 3.1025 | 4 | 304 | 18.4 | 9.97  |
| 311 | 2.63548 | 0.0  | 9.90  | 0 | 0.5440 | 4.973 | 37.8 | 2.5194 | 4 | 304 | 18.4 | 12.64 |
| 312 | 0.79041 | 0.0  | 9.90  | 0 | 0.5440 | 6.122 | 52.8 | 2.6403 | 4 | 304 | 18.4 | 5.98  |
| 313 | 0.26169 | 0.0  | 9.90  | 0 | 0.5440 | 6.023 | 90.4 | 2.8340 | 4 | 304 | 18.4 | 11.72 |
| 314 | 0.26938 | 0.0  | 9.90  | 0 | 0.5440 | 6.266 | 82.8 | 3.2628 | 4 | 304 | 18.4 | 7.90  |
| 315 | 0.36920 | 0.0  | 9.90  | 0 | 0.5440 | 6.567 | 87.3 | 3.6023 | 4 | 304 | 18.4 | 9.28  |
| 316 | 0.25356 | 0.0  | 9.90  | 0 | 0.5440 | 5.705 | 77.7 | 3.9450 | 4 | 304 | 18.4 | 11.50 |
| 317 | 0.31827 | 0.0  | 9.90  | 0 | 0.5440 | 5.914 | 83.2 | 3.9986 | 4 | 304 | 18.4 | 18.33 |



|     |         |      |       |   |        |       |      |         |    |     |      |       |
|-----|---------|------|-------|---|--------|-------|------|---------|----|-----|------|-------|
| 318 | 0.24522 | 0.0  | 9.90  | 0 | 0.5440 | 5.782 | 71.7 | 4.0317  | 4  | 304 | 18.4 | 15.94 |
| 319 | 0.40202 | 0.0  | 9.90  | 0 | 0.5440 | 6.382 | 67.2 | 3.5325  | 4  | 304 | 18.4 | 10.36 |
| 320 | 0.47547 | 0.0  | 9.90  | 0 | 0.5440 | 6.113 | 58.8 | 4.0019  | 4  | 304 | 18.4 | 12.73 |
| 321 | 0.16760 | 0.0  | 7.38  | 0 | 0.4930 | 6.426 | 52.3 | 4.5404  | 5  | 287 | 19.6 | 7.20  |
| 322 | 0.18159 | 0.0  | 7.38  | 0 | 0.4930 | 6.376 | 54.3 | 4.5404  | 5  | 287 | 19.6 | 6.87  |
| 323 | 0.35114 | 0.0  | 7.38  | 0 | 0.4930 | 6.041 | 49.9 | 4.7211  | 5  | 287 | 19.6 | 7.70  |
| 324 | 0.28392 | 0.0  | 7.38  | 0 | 0.4930 | 5.708 | 74.3 | 4.7211  | 5  | 287 | 19.6 | 11.74 |
| 325 | 0.34109 | 0.0  | 7.38  | 0 | 0.4930 | 6.415 | 40.1 | 4.7211  | 5  | 287 | 19.6 | 6.12  |
| 326 | 0.19186 | 0.0  | 7.38  | 0 | 0.4930 | 6.431 | 14.7 | 5.4159  | 5  | 287 | 19.6 | 5.08  |
| 327 | 0.30347 | 0.0  | 7.38  | 0 | 0.4930 | 6.312 | 28.9 | 5.4159  | 5  | 287 | 19.6 | 6.15  |
| 328 | 0.24103 | 0.0  | 7.38  | 0 | 0.4930 | 6.083 | 43.7 | 5.4159  | 5  | 287 | 19.6 | 12.79 |
| 329 | 0.06617 | 0.0  | 3.24  | 0 | 0.4600 | 5.868 | 25.8 | 5.2146  | 4  | 430 | 16.9 | 9.97  |
| 330 | 0.06724 | 0.0  | 3.24  | 0 | 0.4600 | 6.333 | 17.2 | 5.2146  | 4  | 430 | 16.9 | 7.34  |
| 331 | 0.04544 | 0.0  | 3.24  | 0 | 0.4600 | 6.144 | 32.2 | 5.8736  | 4  | 430 | 16.9 | 9.09  |
| 332 | 0.05023 | 35.0 | 6.06  | 0 | 0.4379 | 5.706 | 28.4 | 6.6407  | 1  | 304 | 16.9 | 12.43 |
| 333 | 0.03466 | 35.0 | 6.06  | 0 | 0.4379 | 6.031 | 23.3 | 6.6407  | 1  | 304 | 16.9 | 7.83  |
| 334 | 0.05083 | 0.0  | 5.19  | 0 | 0.5150 | 6.316 | 38.1 | 6.4584  | 5  | 224 | 20.2 | 5.68  |
| 335 | 0.03738 | 0.0  | 5.19  | 0 | 0.5150 | 6.310 | 38.5 | 6.4584  | 5  | 224 | 20.2 | 6.75  |
| 336 | 0.03961 | 0.0  | 5.19  | 0 | 0.5150 | 6.037 | 34.5 | 5.9853  | 5  | 224 | 20.2 | 8.01  |
| 337 | 0.03427 | 0.0  | 5.19  | 0 | 0.5150 | 5.869 | 46.3 | 5.2311  | 5  | 224 | 20.2 | 9.80  |
| 338 | 0.03041 | 0.0  | 5.19  | 0 | 0.5150 | 5.895 | 59.6 | 5.6150  | 5  | 224 | 20.2 | 10.56 |
| 339 | 0.03306 | 0.0  | 5.19  | 0 | 0.5150 | 6.059 | 37.3 | 4.8122  | 5  | 224 | 20.2 | 8.51  |
| 340 | 0.05497 | 0.0  | 5.19  | 0 | 0.5150 | 5.985 | 45.4 | 4.8122  | 5  | 224 | 20.2 | 9.74  |
| 341 | 0.06151 | 0.0  | 5.19  | 0 | 0.5150 | 5.968 | 58.5 | 4.8122  | 5  | 224 | 20.2 | 9.29  |
| 342 | 0.01301 | 35.0 | 1.52  | 0 | 0.4420 | 7.241 | 49.3 | 7.0379  | 1  | 284 | 15.5 | 5.49  |
| 343 | 0.02498 | 0.0  | 1.89  | 0 | 0.5180 | 6.540 | 59.7 | 6.2669  | 1  | 422 | 15.9 | 8.65  |
| 344 | 0.02543 | 55.0 | 3.78  | 0 | 0.4840 | 6.696 | 56.4 | 5.7321  | 5  | 370 | 17.6 | 7.18  |
| 345 | 0.03049 | 55.0 | 3.78  | 0 | 0.4840 | 6.874 | 28.1 | 6.4654  | 5  | 370 | 17.6 | 4.61  |
| 346 | 0.03113 | 0.0  | 4.39  | 0 | 0.4420 | 6.014 | 48.5 | 8.0136  | 3  | 352 | 18.8 | 10.53 |
| 347 | 0.06162 | 0.0  | 4.39  | 0 | 0.4420 | 5.898 | 52.3 | 8.0136  | 3  | 352 | 18.8 | 12.67 |
| 348 | 0.01870 | 85.0 | 4.15  | 0 | 0.4290 | 6.516 | 27.7 | 8.5353  | 4  | 351 | 17.9 | 6.36  |
| 349 | 0.01501 | 80.0 | 2.01  | 0 | 0.4350 | 6.635 | 29.7 | 8.3440  | 4  | 280 | 17.0 | 5.99  |
| 350 | 0.02899 | 40.0 | 1.25  | 0 | 0.4290 | 6.939 | 34.5 | 8.7921  | 1  | 335 | 19.7 | 5.89  |
| 351 | 0.06211 | 40.0 | 1.25  | 0 | 0.4290 | 6.490 | 44.4 | 8.7921  | 1  | 335 | 19.7 | 5.98  |
| 352 | 0.07950 | 60.0 | 1.69  | 0 | 0.4110 | 6.579 | 35.9 | 10.7103 | 4  | 411 | 18.3 | 5.49  |
| 353 | 0.07244 | 60.0 | 1.69  | 0 | 0.4110 | 5.884 | 18.5 | 10.7103 | 4  | 411 | 18.3 | 7.79  |
| 354 | 0.01709 | 90.0 | 2.02  | 0 | 0.4100 | 6.728 | 36.1 | 12.1265 | 5  | 187 | 17.0 | 4.50  |
| 355 | 0.04301 | 80.0 | 1.91  | 0 | 0.4130 | 5.663 | 21.9 | 10.5857 | 4  | 334 | 22.0 | 8.05  |
| 356 | 0.10659 | 80.0 | 1.91  | 0 | 0.4130 | 5.936 | 19.5 | 10.5857 | 4  | 334 | 22.0 | 5.57  |
| 357 | 8.98296 | 0.0  | 18.10 | 1 | 0.7700 | 6.212 | 97.4 | 2.1222  | 24 | 666 | 20.2 | 17.60 |
| 358 | 3.84970 | 0.0  | 18.10 | 1 | 0.7700 | 6.395 | 91.0 | 2.5052  | 24 | 666 | 20.2 | 13.27 |
| 359 | 5.20177 | 0.0  | 18.10 | 1 | 0.7700 | 6.127 | 83.4 | 2.7227  | 24 | 666 | 20.2 | 11.48 |
| 360 | 4.26131 | 0.0  | 18.10 | 0 | 0.7700 | 6.112 | 81.3 | 2.5091  | 24 | 666 | 20.2 | 12.67 |

|     |          |     |       |   |        |       |       |        |    |     |      |       |
|-----|----------|-----|-------|---|--------|-------|-------|--------|----|-----|------|-------|
| 361 | 4.54192  | 0.0 | 18.10 | 0 | 0.7700 | 6.398 | 88.0  | 2.5182 | 24 | 666 | 20.2 | 7.79  |
| 362 | 3.83684  | 0.0 | 18.10 | 0 | 0.7700 | 6.251 | 91.1  | 2.2955 | 24 | 666 | 20.2 | 14.19 |
| 363 | 3.67822  | 0.0 | 18.10 | 0 | 0.7700 | 5.362 | 96.2  | 2.1036 | 24 | 666 | 20.2 | 10.19 |
| 364 | 4.22239  | 0.0 | 18.10 | 1 | 0.7700 | 5.803 | 89.0  | 1.9047 | 24 | 666 | 20.2 | 14.64 |
| 365 | 3.47428  | 0.0 | 18.10 | 1 | 0.7180 | 8.780 | 82.9  | 1.9047 | 24 | 666 | 20.2 | 5.29  |
| 366 | 4.55587  | 0.0 | 18.10 | 0 | 0.7180 | 3.561 | 87.9  | 1.6132 | 24 | 666 | 20.2 | 7.12  |
| 367 | 3.69695  | 0.0 | 18.10 | 0 | 0.7180 | 4.963 | 91.4  | 1.7523 | 24 | 666 | 20.2 | 14.00 |
| 368 | 13.52220 | 0.0 | 18.10 | 0 | 0.6310 | 3.863 | 100.0 | 1.5106 | 24 | 666 | 20.2 | 13.33 |
| 369 | 4.89822  | 0.0 | 18.10 | 0 | 0.6310 | 4.970 | 100.0 | 1.3325 | 24 | 666 | 20.2 | 3.26  |
| 370 | 5.66998  | 0.0 | 18.10 | 1 | 0.6310 | 6.683 | 96.8  | 1.3567 | 24 | 666 | 20.2 | 3.73  |
| 371 | 6.53876  | 0.0 | 18.10 | 1 | 0.6310 | 7.016 | 97.5  | 1.2024 | 24 | 666 | 20.2 | 2.96  |
| 372 | 9.23230  | 0.0 | 18.10 | 0 | 0.6310 | 6.216 | 100.0 | 1.1691 | 24 | 666 | 20.2 | 9.53  |
| 373 | 8.26725  | 0.0 | 18.10 | 1 | 0.6680 | 5.875 | 89.6  | 1.1296 | 24 | 666 | 20.2 | 8.88  |
| 374 | 11.10810 | 0.0 | 18.10 | 0 | 0.6680 | 4.906 | 100.0 | 1.1742 | 24 | 666 | 20.2 | 34.77 |
| 375 | 18.49820 | 0.0 | 18.10 | 0 | 0.6680 | 4.138 | 100.0 | 1.1370 | 24 | 666 | 20.2 | 37.97 |
| 376 | 19.60910 | 0.0 | 18.10 | 0 | 0.6710 | 7.313 | 97.9  | 1.3163 | 24 | 666 | 20.2 | 13.44 |
| 377 | 15.28800 | 0.0 | 18.10 | 0 | 0.6710 | 6.649 | 93.3  | 1.3449 | 24 | 666 | 20.2 | 23.24 |
| 378 | 9.82349  | 0.0 | 18.10 | 0 | 0.6710 | 6.794 | 98.8  | 1.3580 | 24 | 666 | 20.2 | 21.24 |
| 379 | 23.64820 | 0.0 | 18.10 | 0 | 0.6710 | 6.380 | 96.2  | 1.3861 | 24 | 666 | 20.2 | 23.69 |
| 380 | 17.86670 | 0.0 | 18.10 | 0 | 0.6710 | 6.223 | 100.0 | 1.3861 | 24 | 666 | 20.2 | 21.78 |
| 381 | 88.97620 | 0.0 | 18.10 | 0 | 0.6710 | 6.968 | 91.9  | 1.4165 | 24 | 666 | 20.2 | 17.21 |
| 382 | 15.87440 | 0.0 | 18.10 | 0 | 0.6710 | 6.545 | 99.1  | 1.5192 | 24 | 666 | 20.2 | 21.08 |
| 383 | 9.18702  | 0.0 | 18.10 | 0 | 0.7000 | 5.536 | 100.0 | 1.5804 | 24 | 666 | 20.2 | 23.60 |
| 384 | 7.99248  | 0.0 | 18.10 | 0 | 0.7000 | 5.520 | 100.0 | 1.5331 | 24 | 666 | 20.2 | 24.56 |
| 385 | 20.08490 | 0.0 | 18.10 | 0 | 0.7000 | 4.368 | 91.2  | 1.4395 | 24 | 666 | 20.2 | 30.63 |
| 386 | 16.81180 | 0.0 | 18.10 | 0 | 0.7000 | 5.277 | 98.1  | 1.4261 | 24 | 666 | 20.2 | 30.81 |
| 387 | 24.39380 | 0.0 | 18.10 | 0 | 0.7000 | 4.652 | 100.0 | 1.4672 | 24 | 666 | 20.2 | 28.28 |
| 388 | 22.59710 | 0.0 | 18.10 | 0 | 0.7000 | 5.000 | 89.5  | 1.5184 | 24 | 666 | 20.2 | 31.99 |
| 389 | 14.33370 | 0.0 | 18.10 | 0 | 0.7000 | 4.880 | 100.0 | 1.5895 | 24 | 666 | 20.2 | 30.62 |
| 390 | 8.15174  | 0.0 | 18.10 | 0 | 0.7000 | 5.390 | 98.9  | 1.7281 | 24 | 666 | 20.2 | 20.85 |
| 391 | 6.96215  | 0.0 | 18.10 | 0 | 0.7000 | 5.713 | 97.0  | 1.9265 | 24 | 666 | 20.2 | 17.11 |
| 392 | 5.29305  | 0.0 | 18.10 | 0 | 0.7000 | 6.051 | 82.5  | 2.1678 | 24 | 666 | 20.2 | 18.76 |
| 393 | 11.57790 | 0.0 | 18.10 | 0 | 0.7000 | 5.036 | 97.0  | 1.7700 | 24 | 666 | 20.2 | 25.68 |
| 394 | 8.64476  | 0.0 | 18.10 | 0 | 0.6930 | 6.193 | 92.6  | 1.7912 | 24 | 666 | 20.2 | 15.17 |
| 395 | 13.35980 | 0.0 | 18.10 | 0 | 0.6930 | 5.887 | 94.7  | 1.7821 | 24 | 666 | 20.2 | 16.35 |
| 396 | 8.71675  | 0.0 | 18.10 | 0 | 0.6930 | 6.471 | 98.8  | 1.7257 | 24 | 666 | 20.2 | 17.12 |
| 397 | 5.87205  | 0.0 | 18.10 | 0 | 0.6930 | 6.405 | 96.0  | 1.6768 | 24 | 666 | 20.2 | 19.37 |
| 398 | 7.67202  | 0.0 | 18.10 | 0 | 0.6930 | 5.747 | 98.9  | 1.6334 | 24 | 666 | 20.2 | 19.92 |
| 399 | 38.35180 | 0.0 | 18.10 | 0 | 0.6930 | 5.453 | 100.0 | 1.4896 | 24 | 666 | 20.2 | 30.59 |
| 400 | 9.91655  | 0.0 | 18.10 | 0 | 0.6930 | 5.852 | 77.8  | 1.5004 | 24 | 666 | 20.2 | 29.97 |
| 401 | 25.04610 | 0.0 | 18.10 | 0 | 0.6930 | 5.987 | 100.0 | 1.5888 | 24 | 666 | 20.2 | 26.77 |
| 402 | 14.23620 | 0.0 | 18.10 | 0 | 0.6930 | 6.343 | 100.0 | 1.5741 | 24 | 666 | 20.2 | 20.32 |
| 403 | 9.59571  | 0.0 | 18.10 | 0 | 0.6930 | 6.404 | 100.0 | 1.6390 | 24 | 666 | 20.2 | 20.31 |

|     |          |     |       |   |        |       |       |        |    |     |      |       |
|-----|----------|-----|-------|---|--------|-------|-------|--------|----|-----|------|-------|
| 404 | 24.80170 | 0.0 | 18.10 | 0 | 0.6930 | 5.349 | 96.0  | 1.7028 | 24 | 666 | 20.2 | 19.77 |
| 405 | 41.52920 | 0.0 | 18.10 | 0 | 0.6930 | 5.531 | 85.4  | 1.6074 | 24 | 666 | 20.2 | 27.38 |
| 406 | 67.92080 | 0.0 | 18.10 | 0 | 0.6930 | 5.683 | 100.0 | 1.4254 | 24 | 666 | 20.2 | 22.98 |
| 407 | 20.71620 | 0.0 | 18.10 | 0 | 0.6590 | 4.138 | 100.0 | 1.1781 | 24 | 666 | 20.2 | 23.34 |
| 408 | 11.95110 | 0.0 | 18.10 | 0 | 0.6590 | 5.608 | 100.0 | 1.2852 | 24 | 666 | 20.2 | 12.13 |
| 409 | 7.40389  | 0.0 | 18.10 | 0 | 0.5970 | 5.617 | 97.9  | 1.4547 | 24 | 666 | 20.2 | 26.40 |
| 410 | 14.43830 | 0.0 | 18.10 | 0 | 0.5970 | 6.852 | 100.0 | 1.4655 | 24 | 666 | 20.2 | 19.78 |
| 411 | 51.13580 | 0.0 | 18.10 | 0 | 0.5970 | 5.757 | 100.0 | 1.4130 | 24 | 666 | 20.2 | 10.11 |
| 412 | 14.05070 | 0.0 | 18.10 | 0 | 0.5970 | 6.657 | 100.0 | 1.5275 | 24 | 666 | 20.2 | 21.22 |
| 413 | 18.81100 | 0.0 | 18.10 | 0 | 0.5970 | 4.628 | 100.0 | 1.5539 | 24 | 666 | 20.2 | 34.37 |
| 414 | 28.65580 | 0.0 | 18.10 | 0 | 0.5970 | 5.155 | 100.0 | 1.5894 | 24 | 666 | 20.2 | 20.08 |
| 415 | 45.74610 | 0.0 | 18.10 | 0 | 0.6930 | 4.519 | 100.0 | 1.6582 | 24 | 666 | 20.2 | 36.98 |
| 416 | 18.08460 | 0.0 | 18.10 | 0 | 0.6790 | 6.434 | 100.0 | 1.8347 | 24 | 666 | 20.2 | 29.05 |
| 417 | 10.83420 | 0.0 | 18.10 | 0 | 0.6790 | 6.782 | 90.8  | 1.8195 | 24 | 666 | 20.2 | 25.79 |
| 418 | 25.94060 | 0.0 | 18.10 | 0 | 0.6790 | 5.304 | 89.1  | 1.6475 | 24 | 666 | 20.2 | 26.64 |
| 419 | 73.53410 | 0.0 | 18.10 | 0 | 0.6790 | 5.957 | 100.0 | 1.8026 | 24 | 666 | 20.2 | 20.62 |
| 420 | 11.81230 | 0.0 | 18.10 | 0 | 0.7180 | 6.824 | 76.5  | 1.7940 | 24 | 666 | 20.2 | 22.74 |
| 421 | 11.08740 | 0.0 | 18.10 | 0 | 0.7180 | 6.411 | 100.0 | 1.8589 | 24 | 666 | 20.2 | 15.02 |
| 422 | 7.02259  | 0.0 | 18.10 | 0 | 0.7180 | 6.006 | 95.3  | 1.8746 | 24 | 666 | 20.2 | 15.70 |
| 423 | 12.04820 | 0.0 | 18.10 | 0 | 0.6140 | 5.648 | 87.6  | 1.9512 | 24 | 666 | 20.2 | 14.10 |
| 424 | 7.05042  | 0.0 | 18.10 | 0 | 0.6140 | 6.103 | 85.1  | 2.0218 | 24 | 666 | 20.2 | 23.29 |
| 425 | 8.79212  | 0.0 | 18.10 | 0 | 0.5840 | 5.565 | 70.6  | 2.0635 | 24 | 666 | 20.2 | 17.16 |
| 426 | 15.86030 | 0.0 | 18.10 | 0 | 0.6790 | 5.896 | 95.4  | 1.9096 | 24 | 666 | 20.2 | 24.39 |
| 427 | 12.24720 | 0.0 | 18.10 | 0 | 0.5840 | 5.837 | 59.7  | 1.9976 | 24 | 666 | 20.2 | 15.69 |
| 428 | 37.66190 | 0.0 | 18.10 | 0 | 0.6790 | 6.202 | 78.7  | 1.8629 | 24 | 666 | 20.2 | 14.52 |
| 429 | 7.36711  | 0.0 | 18.10 | 0 | 0.6790 | 6.193 | 78.1  | 1.9356 | 24 | 666 | 20.2 | 21.52 |
| 430 | 9.33889  | 0.0 | 18.10 | 0 | 0.6790 | 6.380 | 95.6  | 1.9682 | 24 | 666 | 20.2 | 24.08 |
| 431 | 8.49213  | 0.0 | 18.10 | 0 | 0.5840 | 6.348 | 86.1  | 2.0527 | 24 | 666 | 20.2 | 17.64 |
| 432 | 10.06230 | 0.0 | 18.10 | 0 | 0.5840 | 6.833 | 94.3  | 2.0882 | 24 | 666 | 20.2 | 19.69 |
| 433 | 6.44405  | 0.0 | 18.10 | 0 | 0.5840 | 6.425 | 74.8  | 2.2004 | 24 | 666 | 20.2 | 12.03 |
| 434 | 5.58107  | 0.0 | 18.10 | 0 | 0.7130 | 6.436 | 87.9  | 2.3158 | 24 | 666 | 20.2 | 16.22 |
| 435 | 13.91340 | 0.0 | 18.10 | 0 | 0.7130 | 6.208 | 95.0  | 2.2222 | 24 | 666 | 20.2 | 15.17 |
| 436 | 11.16040 | 0.0 | 18.10 | 0 | 0.7400 | 6.629 | 94.6  | 2.1247 | 24 | 666 | 20.2 | 23.27 |
| 437 | 14.42080 | 0.0 | 18.10 | 0 | 0.7400 | 6.461 | 93.3  | 2.0026 | 24 | 666 | 20.2 | 18.05 |
| 438 | 15.17720 | 0.0 | 18.10 | 0 | 0.7400 | 6.152 | 100.0 | 1.9142 | 24 | 666 | 20.2 | 26.45 |
| 439 | 13.67810 | 0.0 | 18.10 | 0 | 0.7400 | 5.935 | 87.9  | 1.8206 | 24 | 666 | 20.2 | 34.02 |
| 440 | 9.39063  | 0.0 | 18.10 | 0 | 0.7400 | 5.627 | 93.9  | 1.8172 | 24 | 666 | 20.2 | 22.88 |
| 441 | 22.05110 | 0.0 | 18.10 | 0 | 0.7400 | 5.818 | 92.4  | 1.8662 | 24 | 666 | 20.2 | 22.11 |
| 442 | 9.72418  | 0.0 | 18.10 | 0 | 0.7400 | 6.406 | 97.2  | 2.0651 | 24 | 666 | 20.2 | 19.52 |
| 443 | 5.66637  | 0.0 | 18.10 | 0 | 0.7400 | 6.219 | 100.0 | 2.0048 | 24 | 666 | 20.2 | 16.59 |
| 444 | 9.96654  | 0.0 | 18.10 | 0 | 0.7400 | 6.485 | 100.0 | 1.9784 | 24 | 666 | 20.2 | 18.85 |
| 445 | 12.80230 | 0.0 | 18.10 | 0 | 0.7400 | 5.854 | 96.6  | 1.8956 | 24 | 666 | 20.2 | 23.79 |
| 446 | 10.67180 | 0.0 | 18.10 | 0 | 0.7400 | 6.459 | 94.8  | 1.9879 | 24 | 666 | 20.2 | 23.98 |

|     |          |     |       |   |        |       |      |        |    |     |      |       |
|-----|----------|-----|-------|---|--------|-------|------|--------|----|-----|------|-------|
| 447 | 6.28807  | 0.0 | 18.10 | 0 | 0.7400 | 6.341 | 96.4 | 2.0720 | 24 | 666 | 20.2 | 17.79 |
| 448 | 9.92485  | 0.0 | 18.10 | 0 | 0.7400 | 6.251 | 96.6 | 2.1980 | 24 | 666 | 20.2 | 16.44 |
| 449 | 9.32909  | 0.0 | 18.10 | 0 | 0.7130 | 6.185 | 98.7 | 2.2616 | 24 | 666 | 20.2 | 18.13 |
| 450 | 7.52601  | 0.0 | 18.10 | 0 | 0.7130 | 6.417 | 98.3 | 2.1850 | 24 | 666 | 20.2 | 19.31 |
| 451 | 6.71772  | 0.0 | 18.10 | 0 | 0.7130 | 6.749 | 92.6 | 2.3236 | 24 | 666 | 20.2 | 17.44 |
| 452 | 5.44114  | 0.0 | 18.10 | 0 | 0.7130 | 6.655 | 98.2 | 2.3552 | 24 | 666 | 20.2 | 17.73 |
| 453 | 5.09017  | 0.0 | 18.10 | 0 | 0.7130 | 6.297 | 91.8 | 2.3682 | 24 | 666 | 20.2 | 17.27 |
| 454 | 8.24809  | 0.0 | 18.10 | 0 | 0.7130 | 7.393 | 99.3 | 2.4527 | 24 | 666 | 20.2 | 16.74 |
| 455 | 9.51363  | 0.0 | 18.10 | 0 | 0.7130 | 6.728 | 94.1 | 2.4961 | 24 | 666 | 20.2 | 18.71 |
| 456 | 4.75237  | 0.0 | 18.10 | 0 | 0.7130 | 6.525 | 86.5 | 2.4358 | 24 | 666 | 20.2 | 18.13 |
| 457 | 4.66883  | 0.0 | 18.10 | 0 | 0.7130 | 5.976 | 87.9 | 2.5806 | 24 | 666 | 20.2 | 19.01 |
| 458 | 8.20058  | 0.0 | 18.10 | 0 | 0.7130 | 5.936 | 80.3 | 2.7792 | 24 | 666 | 20.2 | 16.94 |
| 459 | 7.75223  | 0.0 | 18.10 | 0 | 0.7130 | 6.301 | 83.7 | 2.7831 | 24 | 666 | 20.2 | 16.23 |
| 460 | 6.80117  | 0.0 | 18.10 | 0 | 0.7130 | 6.081 | 84.4 | 2.7175 | 24 | 666 | 20.2 | 14.70 |
| 461 | 4.81213  | 0.0 | 18.10 | 0 | 0.7130 | 6.701 | 90.0 | 2.5975 | 24 | 666 | 20.2 | 16.42 |
| 462 | 3.69311  | 0.0 | 18.10 | 0 | 0.7130 | 6.376 | 88.4 | 2.5671 | 24 | 666 | 20.2 | 14.65 |
| 463 | 6.65492  | 0.0 | 18.10 | 0 | 0.7130 | 6.317 | 83.0 | 2.7344 | 24 | 666 | 20.2 | 13.99 |
| 464 | 5.82115  | 0.0 | 18.10 | 0 | 0.7130 | 6.513 | 89.9 | 2.8016 | 24 | 666 | 20.2 | 10.29 |
| 465 | 7.83932  | 0.0 | 18.10 | 0 | 0.6550 | 6.209 | 65.4 | 2.9634 | 24 | 666 | 20.2 | 13.22 |
| 466 | 3.16360  | 0.0 | 18.10 | 0 | 0.6550 | 5.759 | 48.2 | 3.0665 | 24 | 666 | 20.2 | 14.13 |
| 467 | 3.77498  | 0.0 | 18.10 | 0 | 0.6550 | 5.952 | 84.7 | 2.8715 | 24 | 666 | 20.2 | 17.15 |
| 468 | 4.42228  | 0.0 | 18.10 | 0 | 0.5840 | 6.003 | 94.5 | 2.5403 | 24 | 666 | 20.2 | 21.32 |
| 469 | 15.57570 | 0.0 | 18.10 | 0 | 0.5800 | 5.926 | 71.0 | 2.9084 | 24 | 666 | 20.2 | 18.13 |
| 470 | 13.07510 | 0.0 | 18.10 | 0 | 0.5800 | 5.713 | 56.7 | 2.8237 | 24 | 666 | 20.2 | 14.76 |
| 471 | 4.34879  | 0.0 | 18.10 | 0 | 0.5800 | 6.167 | 84.0 | 3.0334 | 24 | 666 | 20.2 | 16.29 |
| 472 | 4.03841  | 0.0 | 18.10 | 0 | 0.5320 | 6.229 | 90.7 | 3.0993 | 24 | 666 | 20.2 | 12.87 |
| 473 | 3.56868  | 0.0 | 18.10 | 0 | 0.5800 | 6.437 | 75.0 | 2.8965 | 24 | 666 | 20.2 | 14.36 |
| 474 | 4.64689  | 0.0 | 18.10 | 0 | 0.6140 | 6.980 | 67.6 | 2.5329 | 24 | 666 | 20.2 | 11.66 |
| 475 | 8.05579  | 0.0 | 18.10 | 0 | 0.5840 | 5.427 | 95.4 | 2.4298 | 24 | 666 | 20.2 | 18.14 |
| 476 | 6.39312  | 0.0 | 18.10 | 0 | 0.5840 | 6.162 | 97.4 | 2.2060 | 24 | 666 | 20.2 | 24.10 |
| 477 | 4.87141  | 0.0 | 18.10 | 0 | 0.6140 | 6.484 | 93.6 | 2.3053 | 24 | 666 | 20.2 | 18.68 |
| 478 | 15.02340 | 0.0 | 18.10 | 0 | 0.6140 | 5.304 | 97.3 | 2.1007 | 24 | 666 | 20.2 | 24.91 |
| 479 | 10.23300 | 0.0 | 18.10 | 0 | 0.6140 | 6.185 | 96.7 | 2.1705 | 24 | 666 | 20.2 | 18.03 |
| 480 | 14.33370 | 0.0 | 18.10 | 0 | 0.6140 | 6.229 | 88.0 | 1.9512 | 24 | 666 | 20.2 | 13.11 |
| 481 | 5.82401  | 0.0 | 18.10 | 0 | 0.5320 | 6.242 | 64.7 | 3.4242 | 24 | 666 | 20.2 | 10.74 |
| 482 | 5.70818  | 0.0 | 18.10 | 0 | 0.5320 | 6.750 | 74.9 | 3.3317 | 24 | 666 | 20.2 | 7.74  |
| 483 | 5.73116  | 0.0 | 18.10 | 0 | 0.5320 | 7.061 | 77.0 | 3.4106 | 24 | 666 | 20.2 | 7.01  |
| 484 | 2.81838  | 0.0 | 18.10 | 0 | 0.5320 | 5.762 | 40.3 | 4.0983 | 24 | 666 | 20.2 | 10.42 |
| 485 | 2.37857  | 0.0 | 18.10 | 0 | 0.5830 | 5.871 | 41.9 | 3.7240 | 24 | 666 | 20.2 | 13.34 |
| 486 | 3.67367  | 0.0 | 18.10 | 0 | 0.5830 | 6.312 | 51.9 | 3.9917 | 24 | 666 | 20.2 | 10.58 |
| 487 | 5.69175  | 0.0 | 18.10 | 0 | 0.5830 | 6.114 | 79.8 | 3.5459 | 24 | 666 | 20.2 | 14.98 |
| 488 | 4.83567  | 0.0 | 18.10 | 0 | 0.5830 | 5.905 | 53.2 | 3.1523 | 24 | 666 | 20.2 | 11.45 |
| 489 | 0.15086  | 0.0 | 27.74 | 0 | 0.6090 | 5.454 | 92.7 | 1.8209 | 4  | 711 | 20.1 | 18.06 |

|     |         |     |       |   |        |       |      |        |   |     |      |       |
|-----|---------|-----|-------|---|--------|-------|------|--------|---|-----|------|-------|
| 490 | 0.18337 | 0.0 | 27.74 | 0 | 0.6090 | 5.414 | 98.3 | 1.7554 | 4 | 711 | 20.1 | 23.97 |
| 491 | 0.20746 | 0.0 | 27.74 | 0 | 0.6090 | 5.093 | 98.0 | 1.8226 | 4 | 711 | 20.1 | 29.68 |
| 492 | 0.10574 | 0.0 | 27.74 | 0 | 0.6090 | 5.983 | 98.8 | 1.8681 | 4 | 711 | 20.1 | 18.07 |
| 493 | 0.11132 | 0.0 | 27.74 | 0 | 0.6090 | 5.983 | 83.5 | 2.1099 | 4 | 711 | 20.1 | 13.35 |
| 494 | 0.17331 | 0.0 | 9.69  | 0 | 0.5850 | 5.707 | 54.0 | 2.3817 | 6 | 391 | 19.2 | 12.01 |
| 495 | 0.27957 | 0.0 | 9.69  | 0 | 0.5850 | 5.926 | 42.6 | 2.3817 | 6 | 391 | 19.2 | 13.59 |
| 496 | 0.17899 | 0.0 | 9.69  | 0 | 0.5850 | 5.670 | 28.8 | 2.7986 | 6 | 391 | 19.2 | 17.60 |
| 497 | 0.28960 | 0.0 | 9.69  | 0 | 0.5850 | 5.390 | 72.9 | 2.7986 | 6 | 391 | 19.2 | 21.14 |
| 498 | 0.26838 | 0.0 | 9.69  | 0 | 0.5850 | 5.794 | 70.6 | 2.8927 | 6 | 391 | 19.2 | 14.10 |
| 499 | 0.23912 | 0.0 | 9.69  | 0 | 0.5850 | 6.019 | 65.3 | 2.4091 | 6 | 391 | 19.2 | 12.92 |
| 500 | 0.17783 | 0.0 | 9.69  | 0 | 0.5850 | 5.569 | 73.5 | 2.3999 | 6 | 391 | 19.2 | 15.10 |
| 501 | 0.22438 | 0.0 | 9.69  | 0 | 0.5850 | 6.027 | 79.7 | 2.4982 | 6 | 391 | 19.2 | 14.33 |
| 502 | 0.06263 | 0.0 | 11.93 | 0 | 0.5730 | 6.593 | 69.1 | 2.4786 | 1 | 273 | 21.0 | 9.67  |
| 503 | 0.04527 | 0.0 | 11.93 | 0 | 0.5730 | 6.120 | 76.7 | 2.2875 | 1 | 273 | 21.0 | 9.08  |
| 504 | 0.06076 | 0.0 | 11.93 | 0 | 0.5730 | 6.976 | 91.0 | 2.1675 | 1 | 273 | 21.0 | 5.64  |
| 505 | 0.10959 | 0.0 | 11.93 | 0 | 0.5730 | 6.794 | 89.3 | 2.3889 | 1 | 273 | 21.0 | 6.48  |
| 506 | 0.04741 | 0.0 | 11.93 | 0 | 0.5730 | 6.030 | 80.8 | 2.5050 | 1 | 273 | 21.0 | 7.88  |

medv

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|----|------|
| 1  | 24.0 |
| 2  | 21.6 |
| 3  | 34.7 |
| 4  | 33.4 |
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| 10 | 18.9 |
| 11 | 15.0 |
| 12 | 18.9 |
| 13 | 21.7 |
| 14 | 20.4 |
| 15 | 18.2 |
| 16 | 19.9 |
| 17 | 23.1 |
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| 276 | 32.0 |
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| 280 | 35.1 |
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| 282 | 35.4 |
| 283 | 46.0 |

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| 311 | 16.1 |
| 312 | 22.1 |
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499 21.2
500 17.5
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502 22.4
503 20.6
504 23.9
505 22.0
506 11.9
```

## BART -

```
library(BART)
```

Loading required package: nlme

Loading required package: survival

```
set.seed(1056)
x <- Boston[, c("chas", "dis", "rad")]
y <- Boston[, "medv"]
train <- sample(1:nrow(Boston), nrow(Boston)/5)
xtrain <- x[-train,]
ytrain <- y[-train]

xtest <- x[train,]
ytest <- y[train]

bartfit <- gbart(xtrain, ytrain, x.test = xtest)
```

```
*****Calling gbart: type=1
*****Data:
data:n,p,np: 405, 3, 101
y1,yn: 1.313086, -10.786914
x1,x[n*p]: 0.000000, 1.000000
xp1,xp[np*p]: 0.000000, 8.000000
*****Number of Trees: 200
*****Number of Cut Points: 1 ... 8
*****burn,nd,thin: 100,1000,1
*****Prior:beta,alpha,tau,nu,lambda,offset: 2,0.95,0.795495,3,14.0348,22.6869
*****sigma: 8.488255
```

```
*****w (weights): 1.000000 ... 1.000000
*****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,3,0
*****printevery: 100
```

MCMC

```
done 0 (out of 1100)
done 100 (out of 1100)
done 200 (out of 1100)
done 300 (out of 1100)
done 400 (out of 1100)
done 500 (out of 1100)
done 600 (out of 1100)
done 700 (out of 1100)
done 800 (out of 1100)
done 900 (out of 1100)
done 1000 (out of 1100)
time: 2s
trcnt,tecnt: 1000,1000
```

```
yhat.bart <- bartfit$yhat.test.mean
mean((ytest - yhat.bart)^2)
```

```
[1] 54.20834
```

## Regression Spline -

```
library(splines)
library(caret)
```

Loading required package: ggplot2

Loading required package: lattice

Attaching package: 'caret'

The following object is masked from 'package:survival':

cluster

```

xr <- Boston[, c("chas", "dis", "rad")]
yr <- Boston$medv

data_spline <- data.frame(xr, medv = yr)

set.seed(1056)

train_control <- trainControl(method = "cv", number = 5)

spline_model <- train(
  medv ~ bs(dis, df = 4) + bs(rad, df = 4) + chas,
  data = data_spline,
  method = "lm",
  trControl = train_control
)

summary(spline_model)

```

Call:

```
lm(formula = .outcome ~ ., data = dat)
```

Residuals:

| Min     | 1Q     | Median | 3Q    | Max    |
|---------|--------|--------|-------|--------|
| -14.301 | -4.912 | -1.829 | 2.475 | 34.279 |

Coefficients:

|                    | Estimate | Std. Error | t value | Pr(> t ) |     |
|--------------------|----------|------------|---------|----------|-----|
| (Intercept)        | 24.2791  | 2.9123     | 8.337   | 7.57e-16 | *** |
| `bs(dis, df = 4)1` | -1.7009  | 2.9745     | -0.572  | 0.56770  |     |
| `bs(dis, df = 4)2` | 9.7627   | 3.5525     | 2.748   | 0.00621  | **  |
| `bs(dis, df = 4)3` | -6.2349  | 6.0084     | -1.038  | 0.29992  |     |
| `bs(dis, df = 4)4` | 6.1341   | 6.1114     | 1.004   | 0.31601  |     |
| `bs(rad, df = 4)1` | -0.3019  | 3.2549     | -0.093  | 0.92613  |     |
| `bs(rad, df = 4)2` | -15.9895 | 8.9259     | -1.791  | 0.07385  | .   |
| `bs(rad, df = 4)3` | 77.1498  | 31.1813    | 2.474   | 0.01369  | *   |
| `bs(rad, df = 4)4` | -8.1661  | 2.0469     | -3.989  | 7.62e-05 | *** |
| chas               | 5.7824   | 1.4621     | 3.955   | 8.77e-05 | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.184 on 496 degrees of freedom

Multiple R-squared: 0.2222, Adjusted R-squared: 0.2081  
F-statistic: 15.74 on 9 and 496 DF, p-value: < 2.2e-16

```
spline_preds <- predict(spline_model, newdata = data_spline)
spline_mse <- mean((y - spline_preds)^2)
spline_mse
```

```
[1] 65.66112
```

## Single Layer Neural Network -

```
library(nnet)
```

Attaching package: 'nnet'

The following object is masked from 'package:BART':

class.ind

```
xn <- Boston[, c("chas", "dis", "rad")]
yn <- Boston$medv
data_nn <- data.frame(xn, medv = yn)

set.seed(1056)

train_control <- trainControl(method = "cv", number = 5)

nn_model <- train(
  medv ~ .,
  data = data_nn,
  method = "nnet",
  trControl = train_control,
  tuneGrid = expand.grid(
    size = c(1, 3, 5, 7),
    decay = c(0, 0.01, 0.1)
  ),
  linout = TRUE,
  trace = FALSE,
  maxit = 500
)
```

```
Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
: There were missing values in resampled performance measures.
```

```
nn_model
```

Neural Network

506 samples

3 predictor

No pre-processing

Resampling: Cross-Validated (5 fold)

Summary of sample sizes: 404, 405, 405, 405, 405

Resampling results across tuning parameters:

| size | decay | RMSE     | Rsquared  | MAE      |
|------|-------|----------|-----------|----------|
| 1    | 0.00  | 9.176128 | NaN       | 6.649015 |
| 1    | 0.01  | 8.357875 | 0.1762496 | 6.031830 |
| 1    | 0.10  | 8.110848 | 0.2214530 | 5.826894 |
| 3    | 0.00  | 8.057829 | 0.2245708 | 5.922300 |
| 3    | 0.01  | 7.806269 | 0.2886720 | 5.678478 |
| 3    | 0.10  | 7.798244 | 0.2833708 | 5.588193 |
| 5    | 0.00  | 8.494369 | 0.1817030 | 6.152953 |
| 5    | 0.01  | 7.863508 | 0.2705120 | 5.664364 |
| 5    | 0.10  | 7.595316 | 0.3350499 | 5.361413 |
| 7    | 0.00  | 7.806647 | 0.2857970 | 5.496782 |
| 7    | 0.01  | 7.604678 | 0.3268778 | 5.451930 |
| 7    | 0.10  | 7.435334 | 0.3568928 | 5.289074 |

RMSE was used to select the optimal model using the smallest value.

The final values used for the model were size = 7 and decay = 0.1.

```
nn_preds <- predict(nn_model, newdata = data_nn)
nn_mse <- mean((y - nn_preds)^2)
nn_mse
```

```
[1] 47.5429
```

1. Explain Q1, Q2, and Q3.

My analysis shows that locational variables such as chas, dis, and rad have a meaningful effect on median house price. Properties near the Charles River, closer to employment centers in

Boston, or with better highway accessibility tend to differ in value, but the relationships are not strictly linear. This is supported by the fact that the more flexible nonlinear models outperformed the regression spline, suggesting that housing prices respond to these locational factors in more complex ways than a simple linear trend can capture.

Among the three models, the single layer neural network performed best with an MSE of 47.54, followed by BART with an MSE of 54.21, while the regression spline performed worst with an MSE of 65.66. The stronger performance of the neural network and BART indicates that nonlinear methods are better suited for modeling the effects of these locational variables on housing prices. Even with only three predictors, the models were able to capture meaningful variation in median home value, demonstrating that location plays an important role in predicting housing prices.

1. Comment on any ethical implications of this work.

There are also ethical implications in using these models. Housing data often reflects historical inequalities in neighborhood development and investment, meaning that variables related to location can indirectly capture socioeconomic disparities. If these predictions were used in lending, pricing, or policy decisions, they could reinforce existing inequities. In addition, models like neural networks and BART are less interpretable, making it harder to explain why certain neighborhoods receive higher or lower predicted values. Because of this, while these models can improve predictive accuracy, they should be applied carefully to avoid perpetuating bias and to ensure fairness and transparency.

1. Highly recommended: Create visualizations of your models or just of the best model.

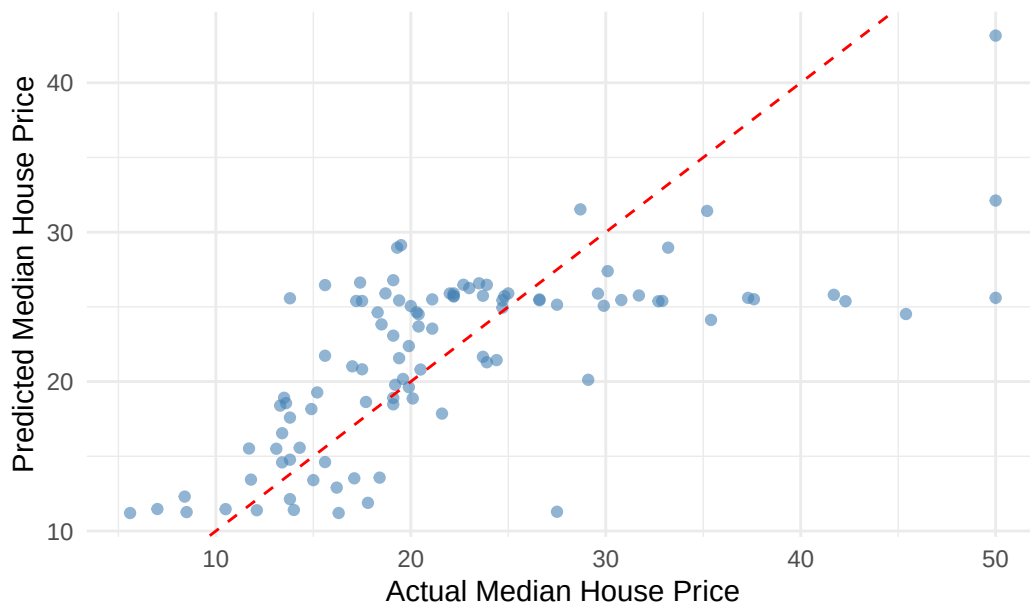
```
library(ggplot2)

nn_preds <- predict(nn_model, newdata = xtest)

plot_data <- data.frame(
  Actual = ytest,
  Predicted = nn_preds
)

ggplot(plot_data, aes(x = Actual, y = Predicted)) +
  geom_point(alpha = 0.6, color = "steelblue") +
  geom_abline(slope = 1, intercept = 0, color = "red", linetype = "dashed") +
  labs(
    title = "Single Layer Neural Network: Predicted vs Actual Mean Housing Price",
    x = "Actual Median House Price",
    y = "Predicted Median House Price"
  ) +
  theme_minimal()
```

## Single Layer Neural Network: Predicted vs Actual Mean Housin



### What teams need to do

1. Answer a team research question.

```
library(MASS)
```

Attaching package: 'MASS'

The following object is masked from 'package:ISLR2':

Boston

```
library(factoextra)
```

Welcome to factoextra!

Want to learn more? See two factoextra-related books at <https://www.datanovia.com/en/product/guide-to-principal-component-methods-in-r/>

```
xk <- Boston[, c("chas", "dis", "rad", "medv")]

x_scaled <- scale(xk)

pca_fit <- prcomp(x_scaled, center = TRUE, scale. = TRUE)

summary(pca_fit)
```

Importance of components:

|                        | PC1    | PC2    | PC3    | PC4    |
|------------------------|--------|--------|--------|--------|
| Standard deviation     | 1.3272 | 1.0537 | 0.8064 | 0.6914 |
| Proportion of Variance | 0.4404 | 0.2776 | 0.1626 | 0.1195 |
| Cumulative Proportion  | 0.4404 | 0.7179 | 0.8805 | 1.0000 |

```
pca_scores <- pca_fit$x[, 1:2]

set.seed(1056)

kfit <- kmeans(pca_scores, centers = 3, nstart = 25)

head(kfit$cluster)
```

```
1 2 3 4 5 6
1 1 1 1 1 1
```

```
kfit$size
```

```
[1] 333 35 138
```

```
kfit$centers
```

|   | PC1        | PC2        |
|---|------------|------------|
| 1 | 0.7195636  | -0.3018794 |
| 2 | 0.3252049  | 3.5350487  |
| 3 | -1.8188178 | -0.1681223 |



```
cluster_labels <- kfit$cluster

cluster_means <- tapply(Boston$medv, cluster_labels, mean)
predicted_medv <- cluster_means[cluster_labels]

mse_cluster_pred <- mean((Boston$medv - predicted_medv)^2)
mse_cluster_pred
```

```
[1] 60.87832
```

1. Explain Q1, Q2, and Q3.
2. Comment on any ethical implications of this work.
3. Create at least one visualization that helps tell the overall narrative of the data.

### **Individual contributions**

Each individual must comment on their contributions to the team research question. Only individuals who contribute to the team section (and state their contributions) will get points for the team section.